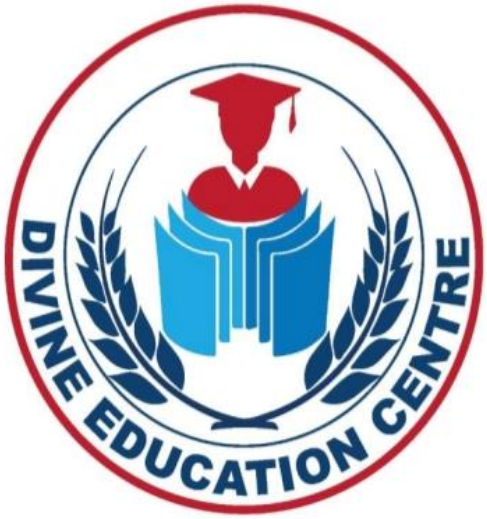
DIVINE EDUCATION CENTRE



MATHEMATICS

LESSON NOTES TERM ONE 2026

0784540287/0751565742

PRIMARY FIVE

P.5 MATHEMATICS LESSON NOTES TERM ONE

TOPICS TO BE COVERED

1. SET CONCEPTS
2. WHOLE NUMBERS
3. OPERATIONS ON WHOLE NUMBERS
4. PATTERNS AND SEQUENCES
5. FRACTIONS (PART A)

TOPIC OUTLINE

THEME : SETS

TOPIC : SET CONCEPTS

- Types of sets
- Parts of a venn diagram
- Listing and finding number of elements in a given set.
- Representing information on a venn diagram
- Subsets
- Probability

SET CONCEPT

Types of sets

A set.

A set is a collection of well-defined elements or members

Review on the following

- Equal sets
- Un equal sets
- Equivalent sets
- Non equivalent sets

Lesson 1

Equal and unequal sets

Equal sets are sets which have exactly the same elements and same number of elements.

The symbol for equal sets is =

Example

Given that set R= {a, b, c} and S= {c ,a, b}

Give the relationship between set R and S

Set R and S are equal sets

OR Set R=S

Unequal sets

Unequal sets are sets with different elements.

Note: For unequal sets ,some elements may be the same but not all elements.

The symbol for unequal sets is ≠

Example

Given that set T= {1,2,3,4,5} and V= { 8,9,4,6}

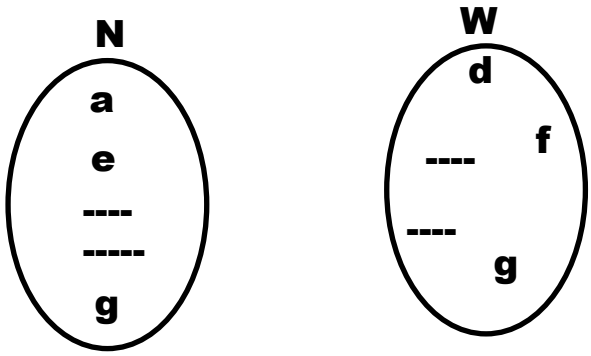
Give the relationship between set T and set V.

Set T and V are unequal sets

Activity

Give the relationship between the following sets

1. Set M={ 0,6,8,9} and set N={8,6,0,9}
2. Set V= {r, u, s, t} and set P={p, o ,t, u, l}
3. Set N={2,3,5,7,11} and set W={1,3,5,7,9}
4. Set R= {b, e, d} and set P={ d, b ,e}
5. Set Q= {x, y, z, p} and set P= {o, p, q, r, s}
6. Fill in the missing elements for set N to be equal to set W



Equivalent and Non equivalent sets.

Equivalent sets

These are sets which have the same number of elements but different elements.

The set symbol for equivalent sets is ⟷

Example

1. Given that set T={b,e,d} and set W= {o,p,q}
   Give the relationship between the two sets

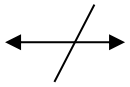
   Set T and W are equivalent sets

   or set T ⟷ set W

2. Give the relationship between set P and set Q.
   P={1,2,3,4} and set Q={a,b,c,d}
   Set P and Q are equivalent sets
   or set P ⟷ set Q

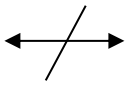
Non equivalent sets

These are sets which have different elements and different number of elements

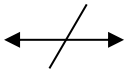
The set symbol for non equivalent sets is 

Example

1. If set R={x,y,z} and set V={ 1,2,3,4,5} . Give the relationship between set R and set V.
   Set R and V are non equivalent sets

   or set R  set V

2. If set U={cow,goat,cat} and set W={plate,fork,cups,saucepan}
   Give the relationship between set U and set W.
   Set U and set W are non equivalent sets

   or or set U  set W

ACTIVITY

Give the relationship between the following sets.

1. Set T={0,2,4,6} and set P={8,9,3,5}
2. Set N={r, u, s, t} and set M={e, c, h ,p, q}
3. Set P={s,o,a,p} and set S={t,w,x,y}
4. Set M={ t,e,a} and set R={ x,y,z}
5. Set A={ 1,2,3,4,5,6} and set B={ 0,7,8,9}
6. State whether equivalent or non equivalent sets

   a) Set N={ b,a,g} and set R={ s,i,t}

Set N and R are_____

b) Set W={bus,taxi,lorry,bicycle} and set M={soda,beer,water}
Set W and M are _____

INTERSECTION SETS (∩)

A set of common elements found in two or more sets.

The symbol for intersection is **∩**

Example I

Given set A = {a, b, c, e, f, g}

B = {b, d, e, f, g}

Find A ∩ B

A ∩ B = {b, e, f, g}

Example II

Set K = {a, e, i, o, u}

B= {a, b, c, d, e, f, g, h, i}

(i) Find (i) K ∩ B = {a, e, i)

(ii) n(K∩B) = 3

Examples III

K = {Odd numbers less than 10}

M = {counting numbers less than 5}

a) List down members of :

(i) Set K

K = {1, 3, 5, 7, 9}

(ii) Set M

M = {1,2, 3,4}

b) Find K∩M = {1,3}

Activity

1. Given set B = {1, 2, 3, 4, 5, 6} and K = {3, 4, 5, 6, 7, 8, 9}

   List members of (ii) A ∩ B

   Find n(A∩B)

2. P = {0, 1, 2, 3, 4, 5}
   Q = {0, 2, 4, 6, 8}
   Find (i) P∩Q
   (ii) n(P∩Q)
3. Given set m = {vowel letters}
   B = {a, b, c, d, e}
   Find (i) M∩B)
   (ii) n(M∩B)
4. Set Q = {Even numbers less than 12}
   B= {4, 6, 8, 9, 10, 12}
   Find (i) Q∩B
   (ii) n(Q∩B)
5. Set H = {Odd numbers less than 13}
   M= {2, 3, 4, 5, 6, 7, 8, 9}
   Find (i) H∩M
   (iii) n(M∩H)

6. Given that set X={ b,o,a,t,c} and set Y={ a,b,c,d,e}
   a) List down the elements common in set X and Y.
   b) How many elements are common in set X and Y?

UNION SET

A union set is a set with all members of given sets.

Note:

Common members of two or more sets are written once.

The symbol for Union **U**

1. Given M = {orange, mangoes, pawpaw}
   K = {tomatoes, peas, pineapples}
   Find MuK = {oranges, mangoes, pawpaw, tomatoes, peas, pineapples)
2. Set H = {4, 6, 8, 9, 10, 12, 14}
   B = {0, 2, 4, 6, 8, 9, 10, 12, 14}
   Find HUB = {0, 2, 4, 6, 8, 9, 10, 12, 14}
   n(HUB) = 9

3. Set W = {whole numbers less than 6}

P= {Odd number less than 13}

Find (i) WUP

(ii) n(WUP)

Soln

W= {0,1,2, 3, 4,5}

P= {1, 3, 5, 7, 9, 11}

WUP = {0,1, 2, 3, 4,5, 7, 9, 11}

n(WUP) = 9

Activities

1. Set M= {1, 2, 3, 4, 5, 6, 7}

   K = {1, 2, 3, 4, 5, 8, 9, 10}

   Find MUK
2. Set H = {4,6,8,9,10} and

   B= {2,3,5,6,7,8}

   Find (i) HUB

   (ii) n(HUB)
3. Set R = {Even numbers less than 14}

   H= {counting numbers less than 12}

   Find (i) RUH

   (ii) n(RUH)
4. Set M = {Natural numbers less than 10}

   B = {1, 3, 5, 7, 9}

   Find (i) MUB

   (ii) n(MUB)
5. Set R = {1, 4, 9,16, 25, 36}

   H = {0, 2, 4, 6, 8, 10, 12}

   Find (i) RUH

   (ii) n(RUH)

UNIVERSAL SET (ξ)

A universal set is a set which contains all elements from which other sets are formed.

The set symbol for universal set is ξ

Example 1

Set M = {1, 2, 3, 4, 5}

Q = {4, 6, 8, 9}

B = {4, 8, 9}

n(ξ) = {1, 2, 3 , 4, 5, 6, 7 , 8 , 9}

DIFFERENCE OF SETS

Is a set of elements that belong to only one set.

Examples

1. Given that set K = {5, 6, 7, 8, 9, 10, 11, 12}

   B = {2, 3, 5, 7, 11, 13}

   Find (i) K – B? (means elements in set K only)

   (ii) B only(means B-

   Soln

   K – B = {6, 8, 9, 10, 12}

   B – K = {2, 3, 13}

2. Given that set E= {Whole numbers less than 8} and

   P = {even numbers between 2 and 14}

   Find (i) P – E

   (ii) E – P

Soln

E = {0, 1, 2, 3, 4, 5, 6, 7}

P= {2, 4,6,8,10,12}

(i) P – E = {8,10,12}

n (P- E) = 1

E – P = {0, 1, 2,3,5, 7}

n (E – P) = 6

3. Given that set Q= {0, 2, 6, 8, 9,10}
M = {1, 2, 3, 4, 6, 11, 13}
Find (i) Q – M
(ii) n (Q – M)
(ii) M – Q
(iii) n (M – Q)
Soln
(i) Q – M = {0, 8, 9, 10}
(ii) n (Q – M) = 4
(iii) M – Q = {1, 3, 11, 13}
n (M – Q) = 4

Activity

1. T = {a, b, c, d, e}
S = {a, e, i, o, u}
Find
(a) T - S
(b) S – T
2. Set H = {all even numbers less than 10}
L = {all odd numbers less than 10}

(a) How many elements are in K – L?

(b) List down all members of L – K?

3. Given that M = {d, n, p, r, t, y} and
N = {c, d, k, m, n, x, v}

(a)List all the members of M and N

(b) Find (i) M – N
(ii) n (M – N)

(c) (iii) N – M
(iv) n (N – M)

4. Given that E = {whole numbers less than 16}
And A = {Odd numbers less than 16}

(a) List the members of E and members of A

E = { } A = { }

(b) Find (i) A – E

(ii) E - A

COMPLEMENT OF SETS

Complement of a set means a set of members that don't belong to a given set.

It is a set of elements outside the mentioned set.

Given A^{I}, this is read as A complement.

Examples 1

Given that: P = {4, 3, 6, 7, 9} and

Q = {1, 2, 3, 5, 7}

Write down members of P^{I} (complement of set P)

P^{I} = {1, 2, 5}

$n(P^{1}) = 3$

Examples II

Given M = {0, 4, 6, 8, 9}

H = {1, 2, 4, 6, 10, 12}

Find m^{1} = {1, 2, 10, 12}

H^{I} = {0, 8, 9}

Examples III

Given N = {4, 6, 8, 9, 10, 12}

R = {0, 2, 4, 6, 8, 10}

Find the compliment of set R

Soln

R^{I} = {9, 12}

(ii) Find N^{I} = {0, 2}

Activity

1. Given A = {1, 2, 3, 4, 5, 7, 9}

   B = {0, 2, 4, 6, 8}

   Find (i) A^1

   (ii) B^1

2. Given set H = {prime members less than 10}

   K = {odd numbers less than 10}

   Find the complement of set K.

3. Set M = {4, 6, 8, 9, 10, 12}

   K = {2, 4, 6, 8, 11, 12}

(a) Find the complement of set M.

(b) Find the complement of set K.

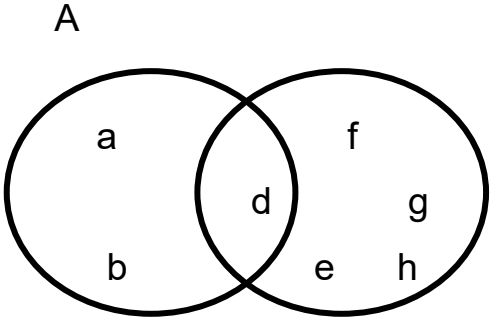
4. Set R = {1, 3, 5, 7, 9,11}

   M= {1, 2, 3, 4, 5, 10, 12}

   Find (i) R^I

   (ii) M^I

5. Below is avenn diagram. Use it to answer questions that follow.

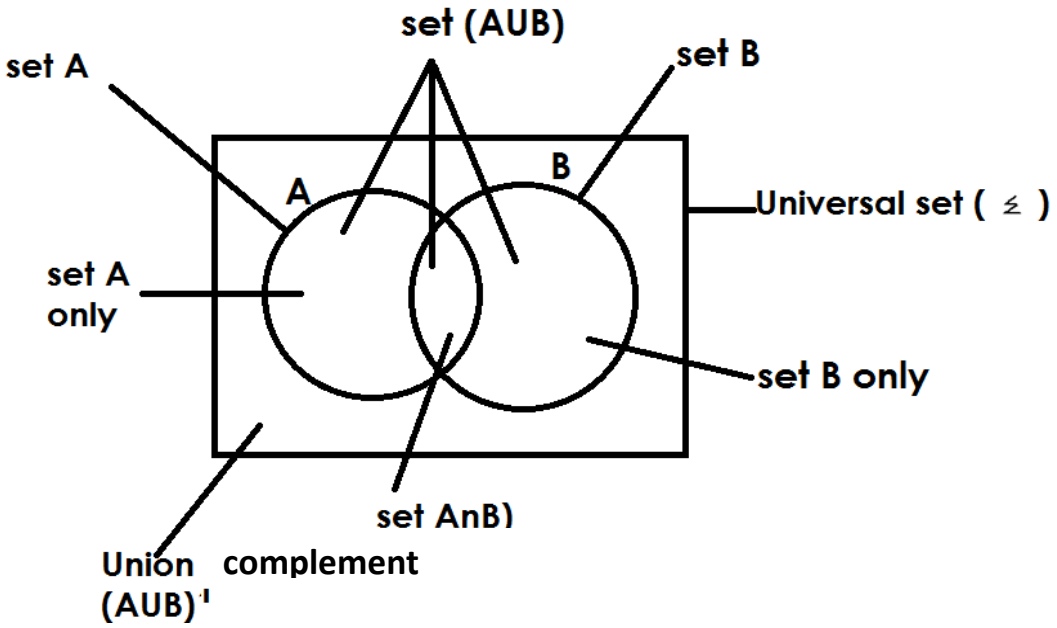


(i) Find (ii) A^1

(ii) $(A \cap B)^1$

(iii) B^1

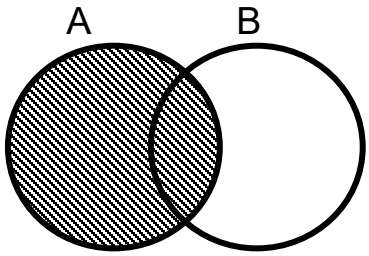
PARTS OF AVENN DIAGRAM



DESCRIBING SHADED AND UNSHADED PARTS OF VENN DIAGRAMS

EXAMPLES

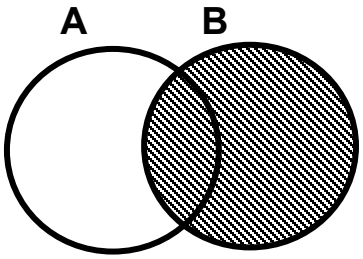
1)



Set A is shaded

Set B only is unshaded

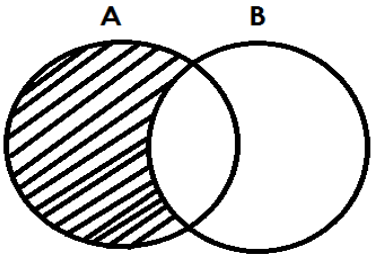
2)



Set B is shaded

Set A only is unshaded

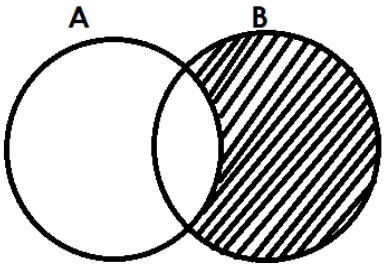
3)



A only / A - B / B^1 is shaded

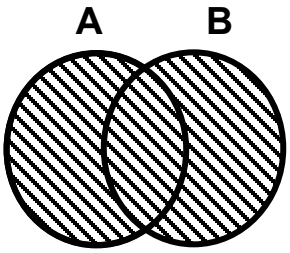
Set B is unshaded

4)

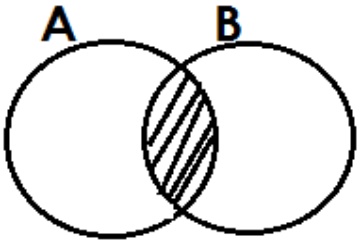


B only / B – A / A^1 is shaded

Set A is unshaded

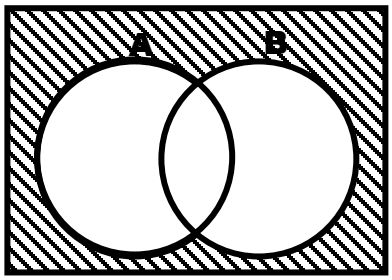
5) 

AUB is shaded

6) 

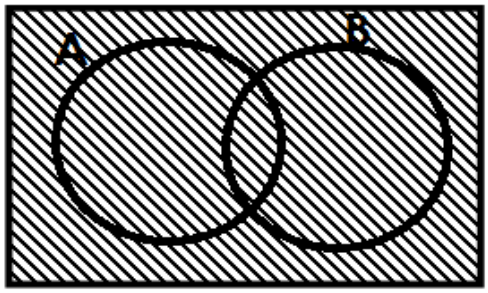
A ∩ B is shaded

$(A \cap B)^1$ is unshaded

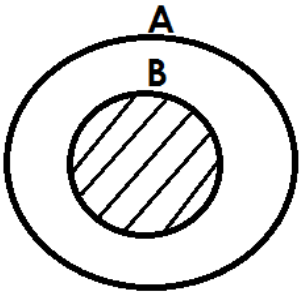
7) 

$(AUB)^1$is shaded

AUB is unshaded)

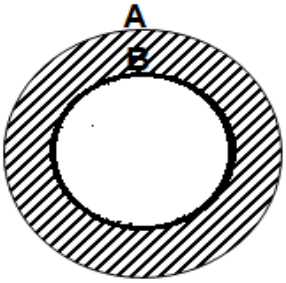
8) 

Universal set is shaded

9) 

Set B /A∩B is shaded

Set B^1 is not shaded

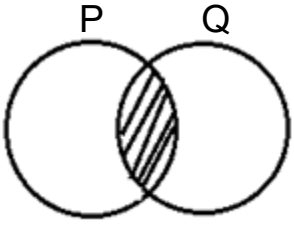
10) 

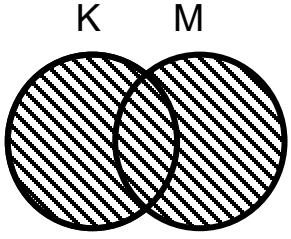
A only / B^I **is shaded**

Set B is not shaded

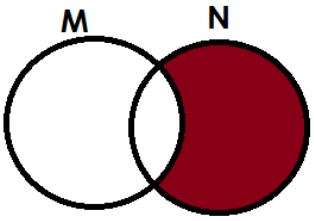
ACTIVITY 1

1. Describe each of the shaded regions in the venn diagrams below.

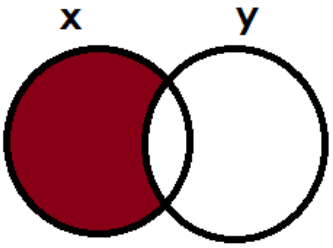
(a) 

(b) 

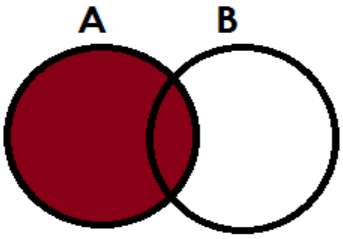
(c)



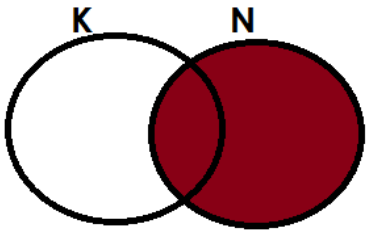
(d)



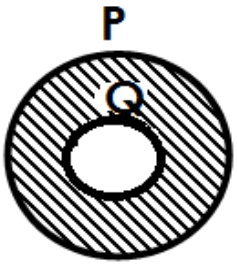
(e)



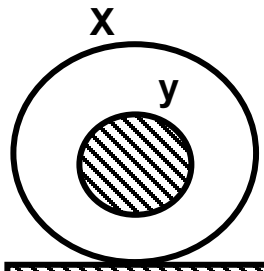
(f)



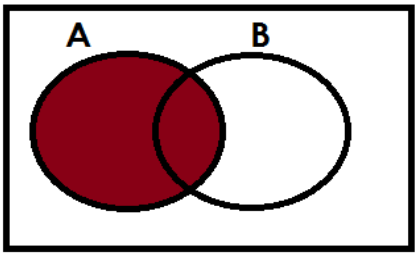
(g)



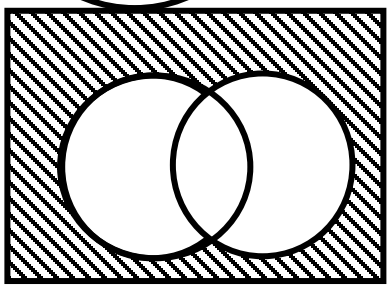
(h)

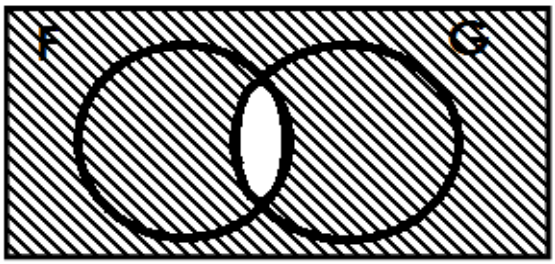


(i)



(j)

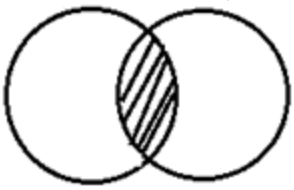




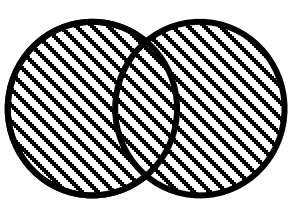
ACTIVITY 2

Describe each of the unshaded regions in the venn diagrams below.

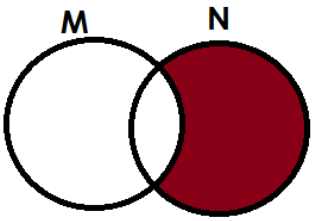
(a)



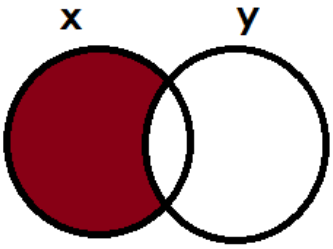
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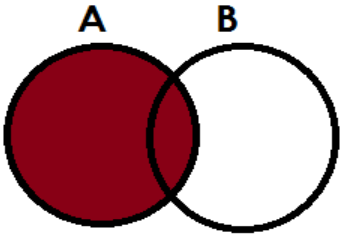
(c)



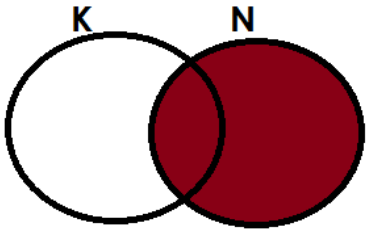
(d)



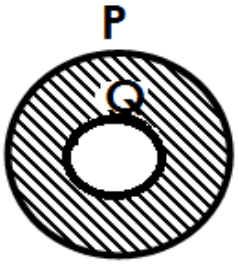
(e)



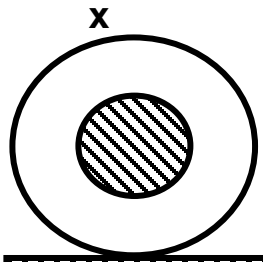
(f)



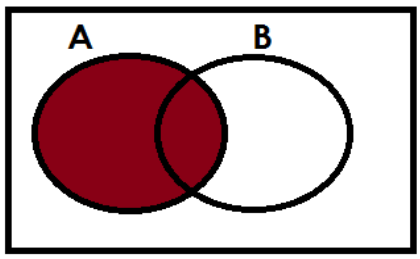
(g)



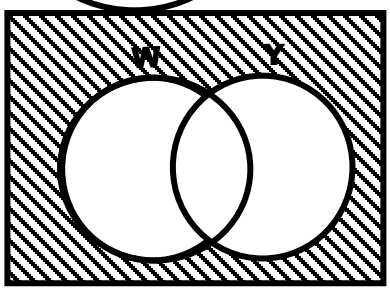
(h)

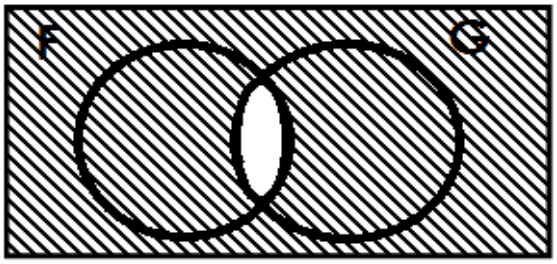


(i)



(j)





LISTING AND FINDING NUMBER OF ELEMENTS IN A GIVEN SET

1. Given that set A = {all factors of 12} and
   B= { even members less than 10}

a. Find

(i) Set A

A = {1, 2, 3, 4, 6, 12}

(ii) Set B

B = {0, 2, 4, 6, 8}

(iii) (A∩B)

A∩B = {2, 4, 6}

(iv) Find n(AUB)

AUB = {1, 2, 3, 4, 6, 12, 0, 8}

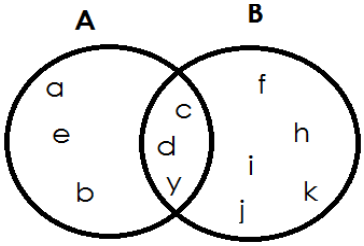
n(AUB)= 8

b. Find n $(A \cap B)^{I}$

$(A \cap B)^{1}$ = {0, 8, 1, 3, 12}

n $(A \cap B)^{1}$ = 5

2. The venn diagram below shows set A and B.



Find

(i) A

Soln

A = {a, b, e, c, d, y}

(ii) (A∩B)1

Soln

(A∩B)1 = {a, e, b, f, h, I, j, k}

n (A∩B) = 8

(ii) (B – A)

soln

B – A = {f, h, I, j, k}

n (B – A) = 5

ACTIVITY

1. Given set P = {All multiples of 4 less than 25}

   Q = {All factors of 16}

   Find (i) P

   (ii) Q

   (iii) P∩Q

   (iv) n(PUQ)

2. Given that E= {whole numbers less than 10}

   P = {2,3,5,7,11}

   Find (i) E

   (ii) n(P)

   (iii) n(E∩P)

   (iv) n(EUP)

3. If set A = {1, 2, 3, 4, 5, 6, 7, 8, 11}

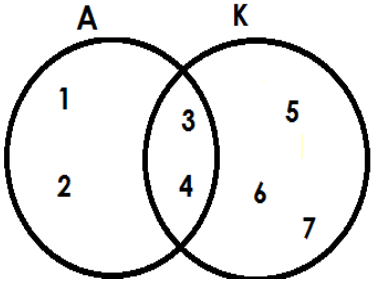
   B = {2, 3, 4, 5, 8, 9, 10, 11}

   Find (i) A∩B

   (ii) n(AUB)

   (iii) n(A – B)

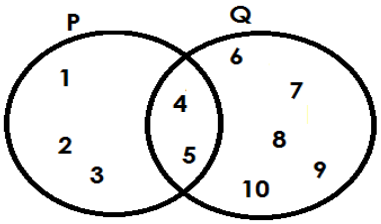
4. Below is set A and K



Find

(i) A
(ii) K
(iii) A∩K
(iv) (A∩K)1
(v) n(AUK)
(vi) $n(A)^1$

5. Use the venn diagram below to answer questions.



Find (i) Set P
(ii) Set Q^1
(iii) n(PUQ)
(iv) $n(P\cap Q)^1$

REPRESENTING INFORMATION ON A VENN DIAGRAMS

A. Given ungrouped information.

Examples

1. Give that set A = {1, 2, 3, 4, 5, 6, 7, 8, 9} and
Set B = { even numbers less than 12}

a. List down the elements in set B

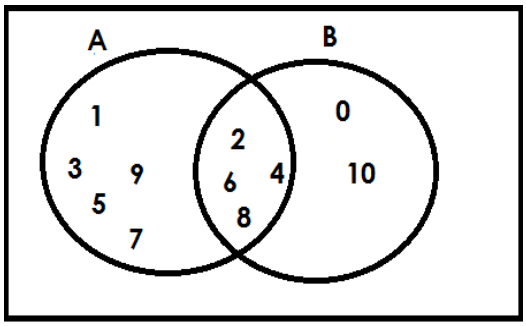
Set B = {0, 2, 4, 6, 8, 10}

b. Represent set A and B on the venn diagram below.

Soln

A = {1, 2, 3, 4, 5, 6, 7, 8, 9}

B = {0, 2, 4, 6, 8, 10}



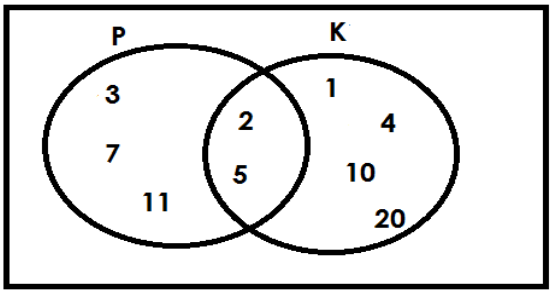
Examples

2. If set P = {prime numbers less than 12} and

   Set K = {all factor of 20}

   P = {2, 3, 5, 7, 11}

   K = {1, 2, 4, 5, 10, 20}



Activity

1. Given that T = {Odd numbers less than 10}
   U = {odd numbers between 6 and 12}
   Show the two sets T and U on a venn diagram.
2. R = {the first five letters of the English alphabets}
   S = {vowels in the English alphabet}
   Represent set R and Set S on the venn diagram.
3. Set K = {all multiples of 4 less than 25} and
   Set T = {even numbers less than 13}
   Represent the above information on the venn diagram.
4. Set M= {all factors of 12} and set N = {all multiples of 3 less than 20}
   Show set M and set N on a venn diagram.

5. If J = {3, 6, 9, 12} and set K = {all even numbers less than 10}

(a) Find set K.

(b) Represent se J and set K on a venn diagram.

6. Given that E = {whole numbers less than 16} and
   A = {odd numbers less than 16}

(a) List the members of E

(b) List the members of set A.

SUBSETS

A subset is a set that can be obtained /formed from any given set.

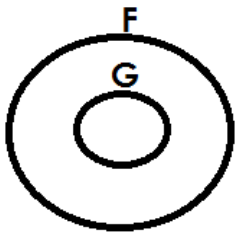
A symbol for subset is $\subseteq$ (is a subset of)

Venn diagram about subsets

Example 1

Draw a venn diagram to show that all girls are female (F)

Soln

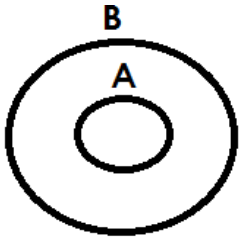


Examples 2

Draw a venn diagram to show that

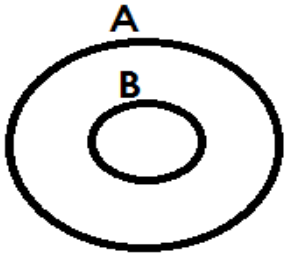
$A = A \cap B$

Soln



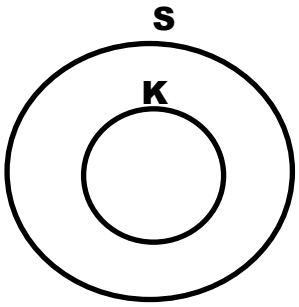
Example 3

Draw a venn diagram to show that A = AUB



Ex

Gi etween the sets below.



Set K is a subset of set S

Or Set K C set S

Activity

1. Draw a venn diagram to show that all hens(h) are birds(b).
2. Draw a venn diagram to show that all boys(B) are males(M).
3. Draw a venn diagram to show that all goats are are animals.
4. Draw a venn diagram to show that P = P∩Q
5. With the help of a venn dram, show that n(K) = n (PUK)

LISTING SUBSETS

Note:

- An empty set is a subset of every set.
- A given set is a subset of itself.
- Equal sets are not subsets of a given set. e.g. {a, b} and {b, a} since they are taken to be the same.

Examples

1. Given that set A = {P}. List all subsets on set A.

   Soln

   { } , {P}

2. If set P = {1, 2}. List all subsets in set T.

   Soln

   { } , {1}, {2}, {1, 2}

3. Set Q = {m, a, n}. List all subsets in set Q.

   Soln

   { }, {m}, {a}, {m, a}, {m, n}, {a, n}, {m, a, n}

Activity

1. List all subsets in sets A. If A = {□}
2. List all subsets in set Q given that Q = [x, y}
3. List all subsets in set B with {△, ○, ▭ }
4. Given that P is {Mary, Amos, Tony}, write down all subsets of P.
5. Given that M = {cow, goat} write all the subsets of set M.
6. List all subsets is set K if K = { }

FINDING NUMBERS OF SUBSETS

No. of subsets = 2^{n}

Where "n" is the number of elements in a given set.

2 shows that number of subsets are in power of 2.

Examples

1. Given that A = { }

Soln

No. of subsets = 2^{n}

2^{0}

1 subset

Note: Any number to power zero is 1

2. If set K = {a,b,c}, find number of subsets in set K.

No. of subsets = 2^{n}

2^{3}

2 x 2 x 2

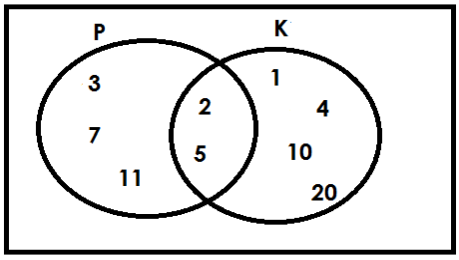
8 subsets

Activity

1. Find the number of subsets of set P, given that P = {1, 2, 3, 4}
2. Set Y = {Toad, Frog} Find the number of subsets in set Y.
3. Find the number of subsets of set X if n(X) = 5.
4. How many subsets are in set W if W = {a}

5. Find the number of subsets in a set with 4 elements.
6. Find the number of subsets in Q given that n(Q) = 1.

7. Use the venn diagram below to answer questions that follow.



a) How many subsets are in set P ∩K?

b) Find number of subsets in (P-K)

PROBABILITY

Probability is the chance that something will happen.

Formula;

Probability = $\frac{n\ (D.\ C)}{n\ (T.C)}$

Where

D.C ⟶ Desired chances

T.C ⟶ Total chances

TOSSING A COIN

A coin has two faces ie

1. Head (H)
2. Tail (T)

Examples

1. A coin was tossed once. What was the probability of a head showing up?

$$\text{Prob} = \frac{n(D.C)}{n(T.C)}$$

D.C = Head

T.C = 1

T.C = Head, Tail

n (T.C) = 2

$$\text{Prob} = \frac{1}{2}$$

Activity

1. Richard tossed a coin once. What is the probability that a head shows up?
2. A girl tossed a coin once. What is the probability that a tail shows up?

ROLLING A DIE

A die has faces numbered; 1, 2, 3, 4, 5, 6

$$\text{Pro} = \frac{n(D.C)}{N(T.C)}$$

Examples

1. A die was rolled once. What is the probability that,

a) An even number appears on top?

D.C = 2, 4, 6

n (D.C) = 3

T.C = 1, 2, 3, 4, 5, 6

n (T.C) = 6

$$\text{Prob} = \frac{3}{6}$$

b) An odd number appears on top.

D.C = 1, 3, 5

n (D.C) = 3

T.C = 1, 2, 3, 4, 5, 6

n (T.C) = 6

Prob = $\frac{3}{6}$

c) A number greater than 2.

D.C = 3, 4, 5, 6

n (D.C) = 4

T.C = 1, 2, 3, 4, 5, 6

n (T.C) = 6

Prob = $\frac{4}{6}$

Activity

1. A die was tossed once. What is the probability of :

(a) a prime number appears on top?

(b) a multiple of 2 appears on top?

(c) a number less than 6 appears top?

(d) a square number appears on top?

(e) a composite number appears on top?

(f) a factor of 4 appears on top

FINDING PROBABILITY GIVEN ITEMS

Examples

1. In a box, there are 4 black pens and 5 red pens. Find the probability of picking a red pen at random.

Soln

n (D.C) = 5

n (T.C) = 4 + 5

= 9

Prob = $\frac{5}{9}$

2. A bag has 6 red shirts and 5 white shirts. If Mukasa picks one shirt at random from the bag. What is the probability of picking a white shirt?

Soln

n (D.C) = 5

n (T.C) = 5 + 6

= 11

Prob = $\frac{5}{11}$

3. In a class, there are 20 boys and 30 girls. What is the probability of picking a boy to be the class captain?

Soln

$$\text{Prob} = \frac{n(D.C)}{n(T.C)}$$

$n(D.C) = 20$

$n(T.C) = 30 + 20$

$= 50$

$\therefore \text{Prob} = \frac{20}{50}$

Activity

1. In a basket, there are 6 ripe mangoes and 3 green one Judith picks one mangoes from the bag without looking. What is the probability that she picks a ripe mango?

2. The letters of the word KAMULI were cut separately and put in a bag. What is the probability of picking at random,

a) a vowel?

b) K?

c) a consonant?

3. In a box these are 2 red pens and 3 black pens. What is the probability that a red pen is picked at random from the box?

4. A bag contains 4 blue places and 5 white plates. What is the probability of picking a white plate from the bag?

5. A box contains fruits, 4 passions fruits, 3 oranges and 2 mangoes. What is the chance that an orange fruit is picked from the box at random?

6. What is the probability of picking a black pen from the tin which contains 3 red pens and 6 black pens?

FINDING PROBABILITY GIVEN DAYS OF THE WEEK

Note:

A week has 7 days. I.e Monday, Tuesday, Wednesday, Thursday, Fridays, Saturday, Sunday.

n (T.C) = 7

1.What is the probability that Musa will go to town on a day which starts with letter S

Soln

Prob = $\frac{\text{n (D.C)}}{\text{n (T.C)}}$

D.C = Saturday, Sunday

n (D.C) = 2

n (T.C) = 7

Prob = $\frac{2}{7}$

Examples

2. Anthony is going to London. What is the probability that is going on a Tuesday?

Soln

D.C = Tuesday

n (D.C) = 1

n (T.C) = 7

Prob = $\frac{1}{7}$

Activity

1. Mr.Kitoko is going for a wedding. What is the probability that he is going on a day which begins with letter T?
2. Agnes is pregnant. If she is expected to give birth next week, what is the probability that she will give birth on Monday?
3. We shall write exams next week. What is the probability that we shall write on a day which begin with letter S?

FINDING PROBABILITY GIVEN MONTH OF THE YEAR

NOTE:

A year has 12 months i.e January, February, March, April, May, June, July, August, September, October, November and December.

Number of total chances is 12.

Example I

We shall go to Kampala next year, what is the probability that we shall go in a month starting with letter J?

D.C = January, June , July

n (D.C) = 3

n (TC) = 12

$$\text{Prob} = \frac{n(D.C)}{n(T.C)}$$

$$= \frac{3}{12}$$

Example II

Matame is wedding next year, what is the probability that she will wed in April.

D.C = April

n (D.C) = 1

n (T.C) = 12

$$\text{Prob} = \frac{n(D.C)}{n(T.C)}$$

$$= \frac{1}{12}$$

Activity

1. Primary leaving examination will be done next year, what is the probability that the examination will be done in the month starting with letter J?
2. What is the probability that Uganda will receive heavy rainfall in the month starting with letter “m”?
3. All government teachers will be paid next year, what is the probability that the payment will be than in the first 3 month of the year.
4. The official election of president will take place next year, what is the probability that will take place in the month starting with letter A?
5. A pregnant mother will give birth next year. What is the probability that it will be in the month of May?

FINDING PROBABILITY INVOLVING FRACTIONS

Example I

The probability that he will pass his examination is $\frac{1}{4}$. What is the probability that he will fail?

Soln

Passing = $\frac{1}{4}$

Failing = $1 - \frac{1}{4}$

= $\frac{4}{4} - \frac{1}{4}$

= $\frac{3}{4}$

Example II

The probability that we shall get food tomorrow is 0.8 what is the probability that we shall not get.

Getting = 0.8

Not getting = $1 - 0.8$

= $\frac{10}{10} - \frac{8}{10}$

= $\frac{2}{10}$

= 0.2

Activity

1. The probability that it will rain today is $\frac{3}{4}$. What is the probability that it will not rain?
2. The probability of passing exams is $\frac{2}{7}$find the probability of failing.
3. The probability that school in Uganda will reopen next month due to covid 19 pandemic is $\frac{1}{3}$. What is the probability that the schools will not open?
4. The probability that Musa will pass an interview is 0.4. What is the probability that he will not pass it?
5. The chance that president Museveni will be voted back in power is 0.6. What is the probability that he will not be voted into power?

TOPIC: WHOLE NUMBERS.

TOPIC OUTLINE

- Forming numerals from digits
- Finding place values and values of whole numbers
- Expanding whole numbers
- Finding the expanded number
- Operations on values of digits in a number
- Operations on values and place values of digits in a number
- Writing figures in words
- Writing words in figures
- Roman and Hindu-Arabic numerals
- Rounding off whole numbers

FORMING NUMERALS FROM DIGITS

Note:

- There is no number that starts with digit zero.
- No commas should be indicted in the number formed.
- When forming the smallest numeral, arrange the given digits in ascending order but without commas.
- When forming the biggest numeral arrange the given digits in descending order but without commas.

Examples I

1. Given digits 2, 3 and 4 form all the three digits numerals

2	3	4
234	324	433

The numbers are 234, 243, 324, 342, 423, 432.

2. Given 4, 0 and 8:

(a) Form the smallest numeral using the above digits.
The smallest numeral = 408

(b) Form the biggest numeral using the above digits.
The biggest numeral = 840

(c) Find the difference between the biggest and smallest numbers formed.

Soln

Biggest = 8 40

Smallest = - 4 0 8

Difference 4 3 2

3. Given the digits 9, 0 and 4

(a) Form all the possible three-digit numerals using the above digits.

Soln

0	4	9
	409	904
	490	940

The numerals are; 409, 490, 904, 940.

4. Given the digits: 4, 0, 2 and 6.

(a) Form the biggest four digit numeral.

Soln

Biggest numeral = 6420

(b) Form the smallest numeral.

Soln

Smallest numeral = 2046

(c) Find the sum of the biggest and the smallest numerals formed.

Biggest numeral ⟶ 6420

Smallest numeral ⟶ 2046

Sum ⟶ +8466

Activity

1. Given the digits 7, 3 and 5.

(a) Form the smallest three-digit number.

(b) Form the biggest three-digit numeral.

(c) Find the sum of the smallest and biggest numerals formed.

2. Given the digits: 9, 1 and 2.

(a) Form the smallest three digits numeral.

(b) Form the biggest three-digit numeral.

(c) Work out the sum of the smallest and the biggest numerals formed.

(d) Find the difference of the biggest and smallest three-digit numerals formed.

(e) Express the smallest three-digit numeral in expanded form.

3. Use 5, 0 and 8 to form all three-digit numerals.

4. Given digits 7, 6 and 9,

a) Form all the three digits numerals.

b) Find the difference of the biggest and the smallest numerals formed.

Finding place values of whole numbers.

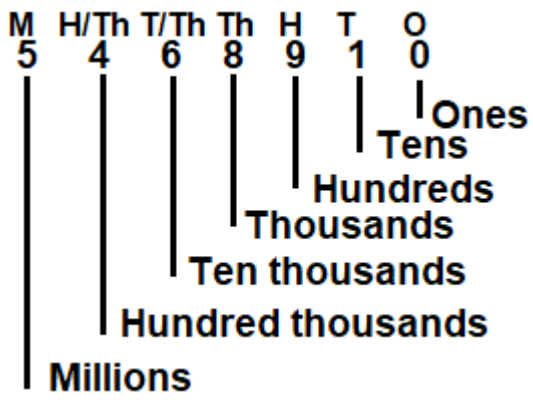
A place value is a position of a digit in a given number.

Example I

Find the place value of each digit in the number below.

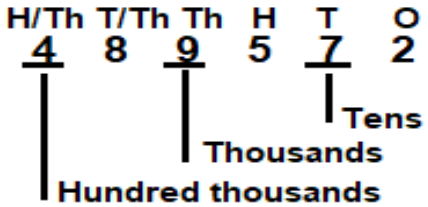
5 4 6 8 9 10

Soln



What is the place value of the underlined digits in 4 8 9 4 7 2

Soln



Activity

1. What is the place valUe of each of the digits in the following.

(a) 5403 (b) 72483 (c) 169542

(d) 623,589 (e) 1689143 (f) 942376

(2) What is the place value of the underlined digits in the following.

(a) 1 4 83 (b) 96432<u>4</u> (c) 7946<u>2</u>

(d) 3<u>8</u>946 (e) 7<u>6</u>421 (f) <u>8</u>46<u>2</u>41

FINDING VALUE OF DIGITS IN WHOLE NUMBERS.

A value is a product of a digit and its place value.

Example I

Find the each of the digits in 4 6 8 2 4 5 9

Soln

M HT TTH TH H T O

4 6 8 2 4 5 9

9 x 1 = 9

5 x 10 = 50

4 x 100 = 400

2 x 1000 = 2000

8 x 10, 000 = 80,000

6 x 100, 0000= 600,000

4 x 1000, 000= 4, 000, 000

Example II

What is the value of the underlined digits in 19 8 4 0 3 6?

M	H/Th	T/Th	Th	H	T	O
1	9	8	4	0	3	6

0 X 100 = 0

8 X 10,000 = 80,000

Activity

1. What is the value of each of the following digits?

(a) 4 03 (b) 97036 (c) 342634 (d) 90786

2. What is the value of the underlined digit.

(a) 3 0 4 (b) 97043 (c) 4 0 3007

3. What is the value of the underlined digit.

(a) 4097 (b) 99043 (c) 102973

EXPANDING WHOLE NUMBERS

(A) **USING PLACES VALUES.**

(a) Expand 7437 using place values.

TH	H	T	O
7	4	3	7

= (7 X 1000) + (4 X 100) + (3 X 10) + (7 X 1)

(b) Express 697,636 in expanded form using place values

HTH	TTH	TH	H	T	O
6	9	7	6	3	6

= (6 X100,000) + (9 X 10,000) + (7 X 1000) + 6 X 100) + (3 X 10) + (6 X 1)

ACTIVITY

1. Expand 9464 using place values.
2. Express 46,367 in expanded from using place values.

3. Expand the following numbers using place values.

(a) 369 (b) 489,630 (c) 346,001 (d) 496,304 (e) 496,346

(B) **USING VALUES**

Examples

1. Expand 4051 using values.

Soln

TH	H	T	O
4	0	5	1

= (4 x 1000) + (0 x 100) + (5 x 10) + (1 x 1)

= 4000 + 0 + 50 + 1

2. Write 817,036 in expanded form using values.

Soln

HTH	TTH	TH	T	H	O
8	1	7	0	3	6

= (8 X 100, 000 + (1 X 10, 000) + (7 X 10, 000) + (0 X 100) + (3 X 10) + (6 X 1)

= 800,000 + 10,000 + 7,000 + 0 + 30 + 6

3. Expand 90760 using values.

Soln

T/Th	TH	H	T	O
9	0	7	6	0

= (9 X 10,000) + (0 X 1000) + (7 X 100) + (6 X 10) + (0 X 1)

= 90,000 + 0 + 700 + 60 + 0

Activity

1. Expand 3049 using values.
2. Express 90347 in expanded form using values.
3. Expand the following numbers.

(a) 304 (b) 4906 (c) 490,763

(d) 396,003 (e) 96,364 (f) 304,304

EXPANDING NUMBERS USING POWERS / INDICES/ EXPONENTS

Examples.

1. Expand 8461 using powers

8	4	6	1
10^3	10^2	10^1	10^0

$= (8 \times 10^3) + (4 \times 10^2) + (6 \times 10^1) + (1 \times 10^0)$

2. Express 52637 in expanded form using exponents.

5	2	6	3	7
10^4	10^3	10^2	10^1	10^0

$= (5 \times 10^4) + (2 \times 10^3) + (6 \times 10^2) + (3 \times 10^1) + (7 \times 10^0)$

3. Expand 40,370 using exponents indices.

4	0,	3	7	0
10^4	10^3	10^2	10^1	10^0

$= (4 \times 10^4) + (0 \times 10^3) + (3 \times 10^2) + (7 \times 10^1) + (0 \times 10^0)$

Activity

1. Expand 304 using powers.
2. Express 46,763 in expanded form using exponents.
3. Expand 503466 using indices.
4. Expand the following numbers using powers.

(a) 30,467 (b) 340,067

(c) 490,363 (d) 963,679

FINDING EXPANDED NUMBER

Examples

1. Express (5 x 100000) + (6 x10000) + (9 x1000) + (4 x 10) as single number.

Soln

= 500,000 + 60,000 + 9,000 + 40

500,000
60,000
9, 000
\+ 40
569,040

2. What number has been expanded to give: 40,000 + 9000+ 800 + 2?

40,000
9,000
800
\+ 2
49,802

3. Write the expanded number in short

= $(2 \times 10^4) + (8 \times 10^3) + (5 \times 10^2) + (3 \times 10^0)$

= (2 x 10 x 10 x 10 x 10) + (8 x 10 x 10 x 10) + (5 x10 x 10) + (3 x 1)

= (2 x 10000) + (8 x 1000) + (5 x 100) + (3 x1)

= 20000 + 8000 + 5 00 + 3

20,000
8,000
500
\+ 3
28,503

4. What number was expanded to give. (3 x 10,000) + (2 x 10)

Soln

30, 000 + 20

30, 000
\+ 20
30,020

5. What number was expanded to give.

$= (3 \times 10^{4}) + (5 \times 10^{2}) + (4 \times 100)$

$= (3 \times 10 \times 10 \times 10 \times 10) + (5 \times 10 \times 10) + (4 \times 1)$

$= (3 \times 10{,}000) + (5 \times 100) + (4 \times 1)$

$= 30{,}000 + 500 + 4$

$$\begin{array}{r} 30{,}000 \\ 500 \\ +\quad 4 \\ \hline 30{,}504 \end{array}$$

Activity

1. What number has been expanded to give.

(a) $(9 \times 1000) + (3 \times 100) + (2 \times 10) + (1 \times 1)$

(b) $(8 \times 100{,}000) + (7 \times 10{,}000) + (2 \times 100) + (2 \times 1)$

(c) $(5 \times 100{,}000) + (6 \times 100) + (4 \times 1)$

2. Write the expanded number in short.

(a) $30{,}000 + 900 + 3$

(b) $9{,}000{,}000 + 700{,}000 + 40$

3. What number was expanded to give.

(a) $(9 \times 10^{3}) + (4 \times 10^{2}) + (2 \times 10^{1}) + (5 \times 10^{0})$

(b) $(6 \times 10^{4}) + (4 \times 10^{2}) + (3 \times 10^{1}) + (0 \times 10^{0})$

(c) $(9 \times 10^{4}) + (3 \times 10^{1}) + (4 \times 10^{0})$

Operations on values of digits in a number

Note:

1. Sum means the answer got after addition
2. Difference means answer got after subtraction
3. Product means answer got after multiplication
4. Quotient means answer got after dividing

Examples

1. Workout the sum of the values of digits 6 and 9 in 6295

   6x1000=6000
   9x10 = 90
   6090

2. Workout the difference between the values of 7 and 8 in 17846
   7x1000=7000
   -8x100 = 800
   6200

3. Workout the product of the values of digits 2 and 9 in 2489
   2x1000=2000
   9x1 = 9

   2000x9=18000

4. Given the numeral 49236, find the quotient of the values of digits 9 and 3 in the
   above numeral
   9x1000=9000
   3x10=30

   $$\frac{9000}{30} = 300$$

Activity

1. Work out the sum of the values of 4 and 6 in 9436
2. Find the sum of the value of 7 and 5 in 5874
3. Work out the difference between the value of 8 and 5 in 24865
4. Work out the product of the value of 4 and value of 6 in 14265
5. Find the quotient of the value of 8 and 2 in 3892
6. Find the difference between the value of 7 and value of 9 in 7298
7. Find the product of the value of 5 and 8 in 5786
8. Work out the quotient of the value of 4 and value of 5 in 14259

Operations on values and place values of digits in a number

Examples

1. Work out the sum of the value of 8 and the place value of 4 in 8645

   8645

   8x1000=8000 8000+10=8010

   tens=10

2. Work out the difference between the place value of 6 and the value of 7 in 26789

   26789 1000 -700=300

   1000

   7x100=700

3. Work out the product of the value of 8 and place value of 3 in 8236

   8236 8000x 10=80,000

   8x1000=8000

   tens=10

4. Find the quotient of the place value of 9 and value of 4 in 1924

   1924

   100 $\frac{100}{4} = 25$

   4x1=4

Activity

1. Work out the sum of the place value of 7 and the value of 5 in 78529
2. Find the difference between the value of 2 and the place value of 8 in 2386
3. Find the quotient of the value of 9 and the place value of 5 in 9452
4. Find the sum of the place value of 6 and value of 9 in 46295
5. Work out the product of the place value of 2 and value of 7 in 1237
6. Work out the product of the value of 8 and place value of 3 in 8359

WRITING FIGURES IN WORDS

Examples

1. Write 36,948 in words.

Soln

Thousand			units		
H	T	O	H	T	O
	3	6	9	4	8

Thirty-six thousand, nine hundred forty-eight.

2. Write 730,609 in words.

Soln.

Thousand			units		
H	T	O	H	T	0
7	3	0	6	0	9

Seven hundred thirty thousand, six hundred nine.

3. Write 1,403,604 in words

Soln

million			Thousand			units		
H	T	O	H	T	O	H	T	o
		1	4	0	3	6	0	4

One million, four hundred three thousand, six hundred four.

Activity

Write the following in words.

(a) 409 (b) 9463

(c) 97,063 (d) 376489

(e) 17,036 (f) 963468

WRITING NUMBER WORDS IN FIGURES

1. Write Twelve thousand four in figures.

Twelve thousand ⟶ 12,000
Four. ⟶ + 4
12,004

2. Write five hundred twenty – one in figure

Five hundred ⟶ 500

Twenty-one ⟶ + 21

521

3. Write four hundred three thousand six hundred twenty in figures.

Four hundred three thousand ⟶ 403,000
Six hundred. ⟶ 600
Twenty. ⟶ + 20
403, 620

4. Write in figures: Five million, four hundred four thousand, seven hundred sixteen

Five million. ⟶ 5,000,000
Five hundred five thousand. ⟶ 505,000
Five hundred. ⟶ 500
Fifteen. ⟶ + 15
5,505, 515

Activity

Write the following in figures

(a) Eleven thousand, six hundred eleven.

(b) Eighteen thousand, five hundred twenty-six.

(c) Ninety – seven thousand, eighteen.

(d) One hundred fifty thousand twenty.

(e) One hundred thousand, one

(f) One hundred one thousand one hundred one

(g) Six million, nine hundred thousand ten

ROMANS NUMERALS

Number

Is the quantity of something.

Numeral

Is a symbol that represents a number.

TYPES OF NUMERALS

1. Hindu Arabic numerals.
2. Roman numerals

HINDU ARABIC NUMERALS

These are numerals that are used in most parts of the world.

Major Hindu Arabic numerals.

0, 1, 2, 3, 4, 5, 6, 7, 8, 9

ROMAN NUMERALS

1. Basic Roman Numeral

Hindu Arabic	**1**	**5**	**10**	**10**	**50**	**100**	**500**	**1000**
Roman Numeral	**1**	**V**	**X**	**X**	**L**	**C**	**C**	**M**

All Roma numerals should be written in capital letters.

2. Repeated Roman numerals.

Roman Numerals got by repeating **I**

2 = II
3 = III

Roman Numerals got by repeating X

20 = XX

30 = XXX

Roman Numerals got by repeating C

200 = CC
300 = CCC

Roman Numerals got by repeating M

200 = MM

3000 = MMM

NB : The above repeated numerals start with digits 2 or 3.

3. Roman Numerals got by adding 5, 50 or 500

NB.

These always start with digits 6, 7 or 8.

Examples

6 = 5 + 1
= VI
7 = 5 + 2
= VII
8 = 5 + 3
= VIII
60 = 50 + 10
= LX
70 = 50 + 20
= LXX
80 = 50 + 30
= LXXX
600 = 500 + 100
= DC
700 = 500 + 200
= DCC
800 = 500 + 300
= DCCC

4. Roman Numerals got by subtracting from 5, 10, 50, 100 or 1000.

NB

These always start with digits 4 and 9.

Examples

4 = 5 – 1
= IV
9 = 10 – 1
= IX
40 = 50 – 10
= XL
90 = 100 – 10
= XC
400 = 500 – 100
= CD
900 = 1000 – 100
= CM

Hindu Arabic and Roman numerals

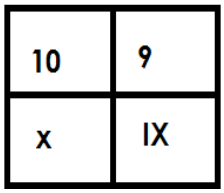
Hindu Arabic	Roman Numeral
1	I
2	II
3	III
4	IV
5	V
6	VI
7	VII
8	VIII
9	IX
10	X
20	XX
30	XXX
40	XL
50	L
60	LX
70	LXX
80	LXXX
90	XC
100	C
200	CC
300	CCC
400	CD
500	D
600	DC
700	DCC
800	DCCC
900	CM
1000	M
2000	MM
3000	MMM

CHANGING HINDU ARABIC NUMERALS TO ROMAN NUMERALS

Examples

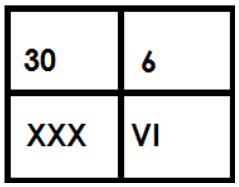
1. Express 19 in Roman Numerals

   Soln

   19 = 

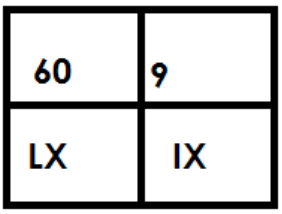
   = XIX

2. Change 36 to Roman numerals.

   36 = 

   = XXXVI

3. Change 69 to Roman numerals

   69 = 

   = LXIX

Activity

Change the following Hindu Arabic numerals to Roman numerals.

(a) 14
(b) 8
(c) 23
(d) 17
(e) 49
(f) 89
(g) 199
(h) 123
(i) 449
(j) 968
(k) 888

CHANGING ROMAN NUMERALS TO HINDU ARABIC NUMERALS

Examples

1. Change XVII to Hindu Arabic Numerals

XVII =

X	VII
10	7

= 17

2. Express XCVIII to Hindu Arabic numeral.

Soln

XCVIII =

XC	VIII
90	8

= 90 + 8

= 98

3. Convert DCCLXXXVIII into Hindu Arabic numeral

Soln

DCCLXXXVIII =

DCC	LXXX	VIII
700	80	8

= 700 + 80 + 8

= 788

Activity

1. Change the following from Roman Numerals to Hindu Arabic numerals.

(a) XI
(b) LVIII
(c) XCIII
(d) CXXIV
(e) CCCLXIII
(f) DCCXX
(g) DCCCLXXXVIII
(h) CMXXII
(I) CMIX

WORD PROBLEMS INVOLVING ROMAN NUMERALS AND HINDU ARABIC NUMERALS

Examples

1. Musoke is 9 years. Write his age in Roman numerals.

Soln

9 = IX

2. Mukuba scored 54 marks in a test. Write score in Roman numerals.

Soln

54 = 50 + 4

LIX

3. There are are XLIV oranges and LVIII apples in a basket. Find the total number of fruits in the basket in Hindu Arabic numerals.

XLIV =	XL	IV
	40	4

+

LVIII =	L	VIII
	50	8

= 44 + 58

= 102 fruits

Activity

1. A basket contains 23 ripe mangoes. Write the number of mangoes in the baskets in Roman numerals.
2. Mukiibi is 49 years. Express his age in Roman numerals.
3. Katongole has XVIII children. Write his number of children in Hindu Arabic numerals
4. There are XLVIII English books in the library. Express the number of books in Hindu Arabic numerals.
5. Malevu was born in MCMXCII. Write his year of birth in Hindu Arabic numerals.

ROUNDING OFF

This is a way of approximating numbers to the required place value.

Rounding off of whole numbers

- In rounding off, consider the digit on the right of the figure in the required place value to either **round up** or **round down**.
- Round down incase the digit on the right of the figure in the required place value is either 0, 1, 2, 3 or 4 by adding zero (0) to the digit in the required place value.
- Round up incase the digit on the right of the figure in the required place value is either 5, 6, 7, 8 or 9 by adding 1 to the digit in the required place value.
- All digits coming on the right of the digit in the required place value will turn into zero (0) when rounding off whole numbers.

Examples

1. Round off 24643 to the nearest hundreds

Soln

Tth	Th	H	T	O
2	4	6	~~4~~	~~3~~
+		~~0~~		
2	4	6	0	0

$\therefore$ 24643 $\approx$ 24600

2. Round off 4896 to the nearest hundredths

Soln

Th	**H**	**T**	**O**
4	8	(9)	(6)
+	~~1~~	0	0
4	9	0	0

$\therefore$ 4896 $\approx$ 4900

3. Round off 68 to the nearest hundreds

H	T	O
0	6	8
+ 1	0	0
1	0	0

$\therefore$ 68 $\frown$ 100

Activity

1. Round off the following to the nearest tens.

(a) 4642
(b) 314
(c) 489
(d) 835
(e) 9

2. Round off the following to the nearest hundreds

(a) 4380
(b) 229
(c) 56
(d) 7625

3. Round off the following to the nearest thousands.

(a) 4687
(b) 96894
(c) 743214
(d) 49684
(e) 14324
(f) 246
(g) 994

OPERATIONS ON WHOLE NUMBERS.

TOPIC OUTLINE

- Addition of whole numbers
- Subtraction of whole numbers
- Multiplication of whole numbers
- Division of whole numbers
- Mixed operations on whole numbers
- Comparing average and total
- Bases
- Finite system

ADDITION OF NUMBERS

Example I

Add: 434126 + 12431

Soln

$$\begin{array}{r} 4\ 3\ 4\ 1\ 2\ 6 \\ +\ 1\ 2\ 4\ 3\ 1 \\ \hline 4\ 4\ 6\ 5\ 5\ 7 \end{array}$$

Example II

Add: 7 6 4 3 2 + 6 8 9 7

Soln

$$\begin{array}{r} 7\ 6\ 4\ 3\ 2 \\ +\quad 6\ 8\ 9\ 7 \\ \hline 7\ 3\ 3\ 2\ 9 \end{array}$$

Example III

Work out : 84399 + 46784

$$\begin{array}{r} 8\ 4\ 3\ 9\ 9 \\ +4\ 6\ 7\ 8\ 4 \\ \hline 13\ 3\ 1\ 1\ 8\ 3 \\ \hline \end{array}$$

Activity

1. Add: 35643 + 12432
2. Add: 432143 + 12344
3. Add: 49986 + 12344
4. Add: 598975 + 146789
5. Add: $\begin{array}{r} 4\ 6\ 9\ 5\ 8\ 9 \\ +3\ 7\ 8\ 9\ 9\ 9 \\ \hline \\ \hline \end{array}$
6. Add: $\begin{array}{r} 1\ 4\ 6\ 5\ 8\ 9 \\ +1\ 3\ 4\ 5\ 9\ 5 \\ \hline \\ \hline \end{array}$
7. Add: 498999 + 110111

WORD PROBLEMS INVOLVING ADDITION OF WHOLE NUMBER

Note:

Sum is the result of addition.

Observe units used in the question.

Example I

In school there are 467 boys and 622 girls. How many pupils are in the school?

Soln

No. of pupils $= 4\ 6\ 7$

$+\ 6\ 2\ 2$

$10\ 8\ 9$ pupils

$\therefore$ 1089 pupils

Examples II

There are 24361 books in the library and 12431, more books were donated to the same library. How many books are there altogether?

Soln

$$\begin{array}{r} 2\ 4\ 3\ 6\ 1 \\ +\ 1\ 2\ 4\ 3\ 1 \\ \hline 3\ 6\ 7\ 9\ 2 \end{array}$$ books.

Activity

1. Add: 48.685 + 89.637
2. What is the sum of 52132 and 93452?
3. Containers A and B have 98456 litres and 45,631 litres of petrol respectively. How much fuel is in the containers altogether?
4. In a region, there are 672,165 women and 489,520 girls. What is the total number of females in the region altogether?
5. A school collected sh.4,586,850 in March and sh 3,950, 980 in April. What was the total of collection in the two months?
6. 853,840ml of juice was consumed in February and 5,950,850 in March. How much juice was consumed in the two months?
7. There are 658,165 pencils in one box and 789,968 in another. What is the total number of pencils in the two boxes?

SUBTRATION OF WHOLE NUMBER

Example I

Subtrac: 4734 – 2412

Soln

$$\begin{array}{r} 4\ 7\ 3\ 4 \\ -\ 2\ 4\ 1\ 2 \\ \hline 2\ 3\ 2\ 2 \end{array}$$

Example II

Work out :

$$\begin{array}{r} 5\ 13\ 11\ 1 \\ 7\not{6}\not{4}\not{2}\ 4 \\ -\quad 1\ 9\ 8\ 6 \\ \hline 7\ 4\ 4\ 3\ 8 \end{array}$$

Example III

Subtract :

$$\begin{array}{r} 4\ 11\ 12 \\ \not{5}\not{2}\ 3\ 3\ 1\ 8\ 6 \\ -\ 1\ 3\ 4\ 5\ 1\ 0\ 2 \\ \hline 3\ 8\ 8\ 8\ 0\ 8\ 4 \end{array}$$

Activity.

Workout the following.

a. 678,028 – 498,762
b. 340,018 – 67,823
c. 567,489,037 – 345,467,254
d. 86421357 – 1358642
e. 6789054 – 3450287
f. 598975 – 146789
g. 432143 – 12344

WORD PROBLEMS INVOLVING SUBTRACTION OF WHOLE NUMBERS.

Note:

Difference is the result after subtracting.

Example I

Subtract 857, 575 from 987628

Soln

$$\begin{array}{r} \quad\quad\ \ 5 \\ 9\ 8\ 7\ \not{6}\ 2\ 8 \\ -\ 8\ 9\ 7\ 5\ 7\ 5 \\ \hline 1\ 3\ 0\ 0\ 5\ 3 \end{array}$$

Example II

Find the difference between 1,650, 922 and 769,866

Soln

$$\begin{array}{r} 15\ 14\ 8\ 1\ 1 \\ \not{1}\not{6}\ \not{5}\not{3}\ 0\ \not{9}\not{2}\ 2 \\ -\ 7\ 6\ 9\ 8\ 6\ 6 \\ \hline 8\ 8\ 1\ 0\ 5\ 6 \end{array}$$

Example III

A farmer had 984636 animals and 462462 died, how many animals remained on the farm.

Remaining animals

$$\begin{array}{r} \not{9}\not{8}\ \not{9}\not{6}\ 3\ 6 \\ -\ 4\ 6\ 2\ 4\ 6\ 2 \\ \hline 5\ 2\ 2\ 1\ 7\ 4 \end{array}$$

Example III

A diary processed 6,500,650 litres of milk and sold 5,650,945 litres. How many litres were left?

A mount of left

$$\begin{array}{r} 5\ 14\ 9\ 9\ \ \ 4 \\ \not{6}\ \not{5}\not{1}\not{0}\not{0}\not{6}\ 5\ \ 0 \\ -\ \ 5\ 6\ 5\ 0\ \ \ 9\ 4\ \ 5 \\ \hline \ \ \ \ 8\ 4\ 9,\ 7\ 0\ \ \ 5\ \text{litres} \end{array}$$

$\therefore$ 849,705 litres were left

Activity

1. Subtract 769,866 from 1,650,922
2. What is the difference between 924568 and 295,877.
3. Subtract: 2,894,052 from 7,014,263
4. Akello's salary sh 1,240,750. It was reduced by sh 89,850. How much does he get now?
5. Out of 2,045,000kg of maize 987,565kg are sold. How many kg remained?
6. The number of cows in a district were 8,004,566 but 595,878 were slaughtered.

   How many cows remained?

7. There were one million chicken on Musa's farm 950,755 were sold on Idd day. How many remained?

MULTIPLICATION OF WHOLE NUMBERS

Example I

1. Multiply: 1 4 2 x 1 2

Soln

$$\begin{array}{r} 1\ 4\ 2 \\ \times\ 1\ 2 \\ \hline 2\ 8\ 4 \\ +1\ 4\ 2\ 0 \\ \hline 1\ 7\ 0\ 4 \end{array}$$

Example II

2. **Workout**

$$\begin{array}{r} 3\ \ 5 \\ \times 1\ \ 2 \\ \hline 7\ \ 0 \\ +3\ 5\ \ 0 \\ \hline 4\ 2\ \ 0 \end{array}$$

3. **Example III**

Multiply 2 4 9 x 32

Soln

$$\begin{array}{r} 2\ \ 4\ \ 9 \\ \times\ \ 3\ \ 2 \\ \hline 4\ \ 9\ \ 8 \\ 7\ 4\ \ 7\ \ 0 \\ \hline 7\ 9\ \ 6\ \ 8 \end{array}$$

Activity

Multiply the following

1. 28 x 11
2. 34 x 12
3. 56 x 23
4. 45 x 25
5. 247 x 23
6. 897 x 42
7. 409 x 73
8. 312 x 25
9. 565 x 20

WORD PROBLEMS INVOLVING MULTIPLICATION OF WHOLE NUMBERS.

Note:

Product is the result of multiplication

4 x 3 = 12

Multiplicand Multiplier Product or multiple

Examples

1. Multiply: 2 2 5 by 14

$$\begin{array}{r} 225 \\ \times\ 14 \\ \hline 900 \\ +2250 \\ \hline 3150 \end{array}$$

2. There are 125 boxes of soap. Each box contains 25 bars of soap. How many bars of soap are there?

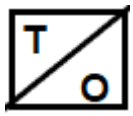
$$\begin{array}{rrrr} & 1 & 2 & 5 \\ \times & & 2 & 5 \\ \hline & 7 & 2 & 5 \\ + \ 2 & 5 & 0 & 0 \\ \hline 3 & 2 & 2 & 5 \end{array}$$ bars of soap

Activity

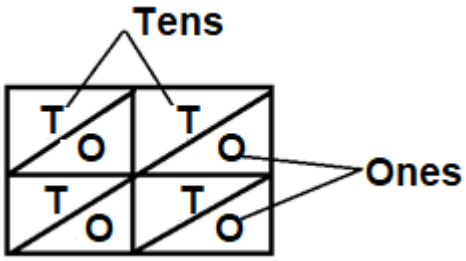
1. Multiply 179 by 19.
2. What is the product of 69 and 76?
3. Find the product of 432 and 63.
4. Multiply 245 by 45.
5. How many pupils are in 33 classrooms if each classroom has 109 pupils?
6. How many nails will 50 boxes hold if each box holds 800 nails?
7. A rectangular garden measure 332 metres by 56 metres. What is the area of the garden in square metre?
8. Fine the area of a rectangular piece of paper whose sides are 42cm by 24cm.

MULTIPICATION OF WNOLE NUMBERS USING LATTICE METHOD.

Note



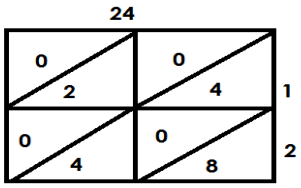
Or



Examples

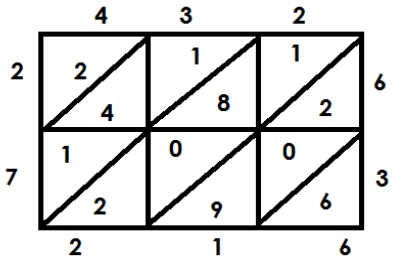
Multiply 2 4 x 12 using lattice method.

Soln



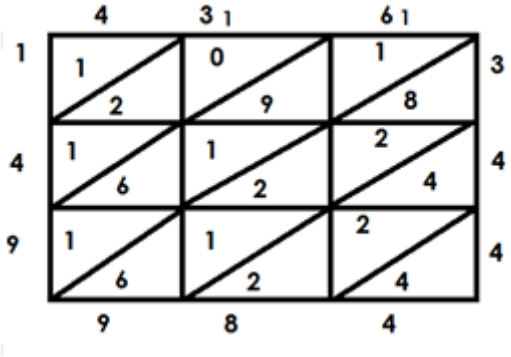
$\therefore$ 24 x 12 = 288

2. Multiply 4 3 2 x 6 3 using lattice method.



∴ 432 x 63 = 27216

3. Multiply: 4 3 6 x 3 4 4



∴ 436 x 344 = 149984

Activity.

<u>Multiply the following using lattice method.</u>

1. 89 x 45
2. 75 x 25
3. 351 x 15
4. 349 x 23
5. 413 x 514
6. 565 x 204

DIVISION OF WHOLE NUMBERS

a) Working of division using repeated subtraction.

Examples

1. Work out 35 ÷ 5 using repeated subtraction

Soln

35 ÷ 5 ⟶ 3 5 - 5 = 30
30 - 5 = 25
25 - 5 = 20
20 - 5 = 15
15 - 5 = 10
10 - 5 = 5
5 - 5 = 0

35 ÷ 5 = 7

2. Work out 49 ÷ 7 using subtraction.

Soln

49 ÷ 7 ⟶ 49 - 7 = 42
42 - 7 = 35
35 - 7 = 28
28 - 7 = 21
21 - 7 = 14
14 - 7 = 7
7 - 7 = 0

49 ÷ 7 = 7

3. Simplify 63 ÷ 7 using repeated subtraction.

Soln

63 ÷ 7 = 63 - 7 = 56
56 - 7 = 49
49 - 7 = 42
42 – 7 = 35
35 – 7 = 28
28 – 7 = 21
14 - 7 = 7
7 – 7 = 0

9 times

∴ 63 ÷ 7 = 9

21 – 7 = 14

Activity

Work out the following using repeated subtraction.

(a) 72 ÷ 9

(b) 16 ÷ 4

(c) 32 ÷ 8

(d) 30 ÷ 5

(e) 54 ÷ 9

(f) 81 ÷ 9

(g) 100 ÷ 20

b) Division of numbers using long division

Examples

Divide $3\overline{)9}$

Soln

3 9

$$\begin{array}{r} 3 \\ 3\overline{)9} \\ 3 \times 3 \; - \; 9 \\ \hline 0 \end{array}$$

$\therefore$ 9 ÷ 3 = 3

Example II

Divide 18 ÷ 2

Soln

$$\begin{array}{r} 0\;9 \\ 2\overline{)1\;8} \\ 0 \times 2 \; = \; 0 \\ \hline 1\;8 \\ 9 \times 2 = -\;1\;8 \\ \hline 0\;0 \end{array}$$

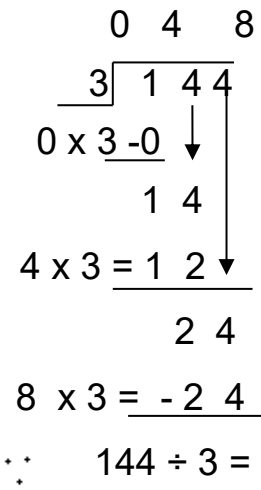
1 ÷ 2 = 0

18 ÷ 2 = 9

$\therefore$ 18 ÷ 2 = 9

Example III

3. Divide 144 ÷ 3



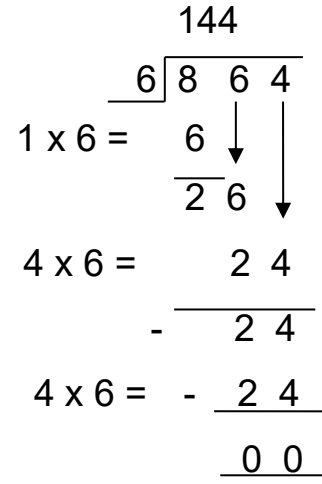
∴ 144 ÷ 3 = 48

1 ÷ 3 = 0

14 ÷ 3 = 4

24 ÷ 3 = 8

4. Divide 864 ÷ 6



8 ÷ 6 = 1

26 ÷ 6 = 4

24 ÷ 6 = 4

<u>Activity 1</u>

<u>Divide the numbers below using long division</u>

1. 24 ÷ 3
2. 48 ÷ 2
3. 15 ÷ 3
4. 72 ÷ 4
5. 144 ÷ 2
6. 400 ÷ 4
7. 1250 ÷ 5

Activity 2

Divide the following

1. $8\overline{)3648}$
2. $6\overline{)5664}$
3. $7\overline{)6538}$
4. $6\overline{)2718}$
5. $6\overline{)1446}$
6. $6\overline{)7268}$

More on division of numbers

Examples

1. Divide 2466 ÷ 9

Soln

$$
\begin{array}{r}
0\,2\,7\,4 \\
9\overline{)\,2\,4\,6\,6} \\
-\,0 \\
\hline
2\,4 \\
-\,1\,8 \\
\hline
6\,6 \\
-\,6\,3 \\
\hline
3\,6 \\
-\,3\,6 \\
\hline
0\,0
\end{array}
$$

0 x 9 = 0

2 x 9 = 18

7 x 9 = 63

4 x 9 = 36

2 ÷ 9 = 0

24 ÷ 9 = 2

68 ÷ 9 = 7

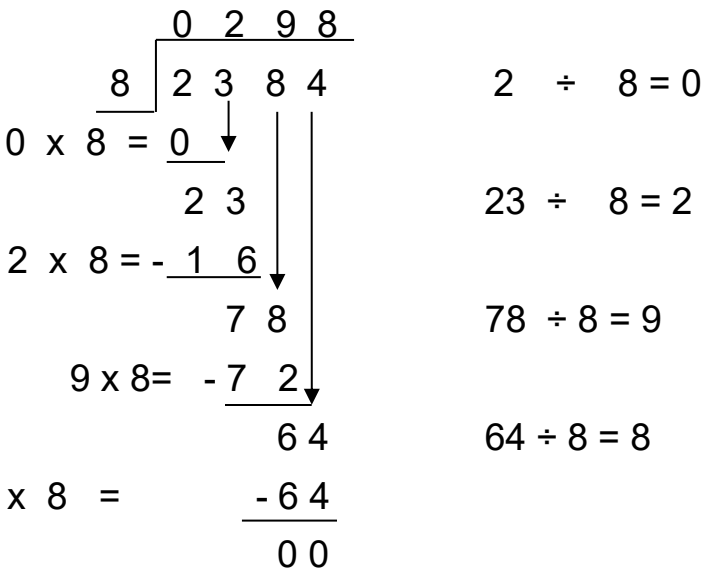
36 ÷ 9 = 4

2466 ÷ 9 = 27

Examples

2. Work out: 2384 ÷ 8

Soln



2384 ÷ 8 = 298

Activity

Divide the following using long division

1. 4578 ÷ 7
2. 5523 ÷ 7
3. 3941 ÷ 7
4. 7268÷ 7

DIVISION BY MULTIPLES OF 10

Examples

1. Divide 2460 ÷ 10

Soln

2460

1Ø

= 246

2. Divide 7630 ÷ 10

Soln

$\frac{7630}{10}$

$= 763$

3. Divide 6200 ÷ 20

$6200 \div 20$

$620 \div 2$

$$\begin{array}{r} 3\ 1\ 0 \\ 2\overline{)\,6\ 2\ 0} \\ 3 \times 2 = 6 \\ \hline 0\ 2 \\ 1 \times 2 = -\ 2 \\ \hline 0\ 0 \\ 0 \times 2 = -\ 0\ 0 \end{array}$$

$\therefore$ $6200 \div 20 = 310$

Activity

Work out

1) $840 \div 10$ 2) $600 \div 30$

3) $1440 \div 4$ 4) $1920 \div 80$

MORE ON DIVISON OF NUMBERS

Division by a two-digit number without a remainder.

Examples

1. Divide 1845 by 15

Soln

1845 ÷ 15

```
          0 1 2 1
       ┌─────────
    15 | 1 8 1 5
0 x 15 =  - 0
          1 8
        - 1 5
          3 1
2 x 15 =  - 3 0
            1 5
1 x 15    - 1 5
            0 0
```

1815 ÷ 15 = 121

2. A lorry carries 25 pieces of timber. How many times will it use to carry 800 pieces of timber?

```
        800 ÷ 25
            0 3 2
        ┌─────────
     25 | 8 0 0
0 x 25 = - 0
           8 0
3 x 25 = - 7 5
             5 0
2 x 25 = - 5 0
           0 0
```

80 ÷ 25 = 3
50 ÷ 25 = 2

∴ The lorry will use 32 times.

Activity

a. 6384 ÷ 14

b. 6840 ÷ 45

c. 7370 ÷ 22

d. 4428 ÷ 36

e. 6840 ÷ 45

f. A carpenter uses 18nails to make I chair. How many chairs will be using 594 nails?

g. A story book of 6500 words has different stories of 50 words each. How many stories are in that story book?

h. One book contains 36 papers. How many such books will be made using 2880 papers?

MIXED OPERATION ON WHOLE NUMBERS.

A) ADDITION AND SUBTRACTION

Note: Re-arrange to add first

Examples

1. Workout 2 – 5 + 9

   (2 + 9) – 8

   = 11 - 5

   = 6

2. Workout: 4 – 6 + 9

   (4 + 9) – 6

   = 13 - 6

   = 7

3. **Work out** 150 – 180 + 400

   (150 + 400) – 180

   1 5 0

   + 4 0 0

   5 5 0

   = 550 – 180

   = 370

Activity

Workout the following

1. 5 – 8 + 9
2. 2 + 9 – 3
3. 9 – 15 + 6
4. 6 – 10 + 7
5. 32 – 40 + 18
6. 24 + 30 – 40
7. 82 – 98 + 35
8. 66 – 80 + 42
9. 80 + 115 – 95

B. **USES OF BODMAS**

AID TO MEMORY

When given questions consisting of 2 or more operation, follow **BODMAS**

Brackets ⟶ (B)

Orders ⟶ (O)

Divide ⟶ (D)

Multiply ⟶ (M)

Add ⟶ (A)

Subtraction ⟶ (S)

Examples

1. Simplify: 5 + 3 x 10

Soln

BODMAS

5 + 3 x 10

= 5 + (3 x 10)

= 5 + 30

= 35

2. Simplify: 5 x 12 ÷ 4

Soln

BODMAS

5 x 12 ÷ 4

= 5 X (12 ÷4)

= 5 X 3

= 15

3. Work out: 18 – (4 x 3) ÷ 6

Soln

BODMAS

18 – 4 x (9 ÷ 3)

18 – (4 x 3)

18 – 12

6

4. Work out 16 ÷ 4 x 3

(16 ÷ 4) x3)

4 x 3

12

Activity

Work out the following

1. 8 + 4 x 5
2. 33 x 2 + 12 ÷ 12
3. $\frac{1}{2}$ of 10 + 15 ÷ 3
4. 14 x 3 – 16
5. (8 – 5) – (3 x 2) + (2 x 2)
6. $\frac{1}{2}$ of 10 + 15 ÷ 5
7. (24 + 16) ÷ 5
8. (8 – 5) – 3 x 2 + (2 x 2)
9. 40 ÷ 8 + 2 – 3
10. 18 – (3 x 8) ÷ 6

COMPARING AVERAGE AND TOTAL

Example I

The average weight of 2 bags is 8kg. Find their total weight.

Soln

Total = Average x number of items

= 2 x 8kg

= 16kg

Example II

The average age of 2 women is 40years. Find their total age.

Soln

Total = Average x number of items

= 40 x 2 = 80 years

Activity

1. The average weight of 2 girls is 40kg find their total weight.
2. The mean age of 6 pupils is 10 years. What is their total age?
3. The average of 12 numbers is 45. Find the total of the 12 numbers.
4. When a box of pencils was shared among 7 pupils each pupil got 3 pencils. How many pencils were in the box?
5. The average score of Opio marks in 4 subjects is 76. What is his total marks?
6. There is an average of 36 pupils in a class if there are 12 classes, how many pupils are in the chair?

BASES

A base is a counting system where a particular counting number is taken to be a limit of its counting system.

Examples of bases

1. Binary base - Base two
2. Ternary base - Base three
3. Quaternary base - Base four
4. Quinary base - Base five
5. Senary base - Base six
6. Septanary base - Base seven
7. Octal base - Base eight
8. Nonary base - Base nine
9. Denary/decimal /nature base - Base ten

GROUPING IN BASE FIVE

Example 1

Group the following in five and write down their number in base five.

(i) 3

3 = | | | 3 ones = 3_{five}

(ii) 8 = 1 group of five, 3 ones = 13five



(iii) 16 = 3groups of fives, I ones = 31 fives



(iv) 28

28 = 1 group of five fives,3 ones =103_{five}



(v) 37

37 = 1 group of five fives, 2 groups of fives 2 ones =121 five



Activity

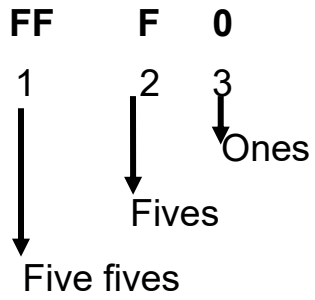
1. Group the following numbers in base five

(a) 4 (b) 7 (c) 17 (d) 29 (e) 42

PLACE VALUE OF BASES

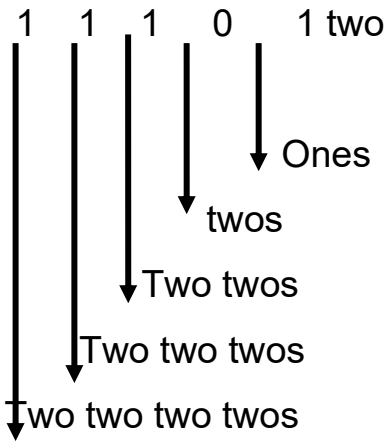
Examples

Find the place value of each digits in 123_{five}.



Examples II

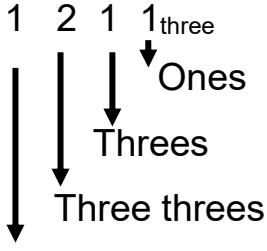
Write the place value of each digit in 1101_{two}



Example III

What is the place value of each digit in 1211_{three}

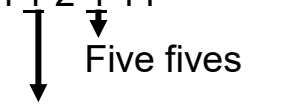
Soln



Three three threes

Examples

Write the place value of the underlined digits in 1 2 4 11



Activity

1. Find the place value of each digit.

(a) 4_{five} (b) 132_{five} (c) 1234_{fives}

2. Find the place value of each digits

(a) 10_{two} (b) 101_{two} (c) 101_{two}

3. Find the place value of each digit

(a) 12_{three} (b) 202_{three} (c) 1202_{three}

4. Write the place value of the underlined digits.

(a) 1 2 4_{five} (b) 1 0 1 $_{two}$ (c) 2 1 1 2 $_{three}$

5. What is the place value of 4 in 2401_{five}.

WRITING BASES IN WORDS

Examples I

1. 123_{five} ⟶ One two three base five.

Example II

4132five ⟶ four one three two base five.

Example III

101_{two} ⟶ One zero one base two.

2102_{three} ⟶ Two one zero two base three

Activity

Write the following numbers in words.

(a)43_{five} (b) 123_{five}

(c) 2142_{five} (d) 1011_{two}

(e)201_{three} (f) 1010_{two}

(g)2102_{three} (h)4232_{five}

EXPANDING BASES USING VALUES OF THE STANDARD BASE

Examples

1. Expand 32_{five} using values.

	2
five	
5^1	5^0

$(3 \times 5^{1}) + (2 \times 5^{0})$

$(3 \times 5) + (2 \times 1)$

$15 + 2$

2. Expand 124_{five} using values.

1 2 4 five ⟶

1	2	4	
5^2	5^1	5^0	five

$(1 \times 5^{2}) + (2 \times 5^{1}) + (4 \times 5^{0})$

$(1 \times 5 \times 5) + (2 \times 5) + (4 \times 1)$

$25 + 10 + 4$

3. Expand 324_{five} using values.

324_{five} ⟶

3	2	4	
5^2	5^1	5^0	five

$(3 \times 5^{2}) + (2 \times 5^{1}) + (4 \times 5^{0})$

$(3 \times 5 \times 5) + (2 \times 5) + (4 \times 1)$

$(3 \times 25) + 10 + 4$

$75 + 10 + 4$

4. Expand 102_{three} using values

102_{three} =

1	0	2	
3^2	3^1	3^0	three

$(1 \times 32) + (0 \times 31) + (2 \times 30)$

$(1 \times 3 \times 3) + (0 \times 3) + (2 \times 1)$

$9 + 0 + 2$

Activity

Expand the following using values.

(a) 43_{five} (b) 123_{five} (c) 432_{five}

(d) 104_{five} (e) 101_{three} (f) 102_{three}

EXPANDING BASES USING POWERS/ EXPONENTS/ INDICES

Examples

Expand 403_{five} using powers

Soln

403_{five} =

4	0	3	five
5^2	5^1	5^0	

$(4 \times 5^2) + (0 \times 5^1) + (3 \times 5^0)$

Expand the following using powers.

(a) 423_{five} =

4	2	3	five
5^2	5^1	5^0	

$(4 \times 5^2) + (2 \times 5^1) + (3 \times 5^0)$

(b) 123_{five}

123_{five} =

1	2	3	five
5^2	5^1	5^0	

$(1 \times 5^2) + (2 \times 5^1) + (3 \times 5^0)$

(c) 102_{three}

102_{three} =

1	0	2	three
3^2	3^1	3^0	

$(1 \times 3^2) + (0 \times 3^1) + (2 \times 3^0)$

Activity

1. Expand the following using p

(a) 124five (b) 243 five (c) 423 five

(d) 104 five (e) 103 four

2. Expand 403 using exponents

EXPANDING BASES USING PLACE VALUES

1. Expand 203five using powers.

2	0	3	five
5^2	5^1	5^0	

$(2 \times 5^2) + (0 \times 5^1) + (3 \times 5^0)$

$(2 \times 5 \times 5) + (0 \times 5) + (3 \times 1)$

2. Expand 403 five using place values.

4	0	3	five
5^2	5^1	5^0	

$(4 \times 5^2) + (0 \times 5^1) + (3 \times 5^0)$

$(4 \times 5 \times 5) + (0 \times 5) + (3 \times 1)$

3. Expand 123_{five} using place values

1	2	3
ff 1	f 2	0 3

$(1 \times 5 \times 5) + (2 \times 5) + (3 \times 1)$

Activity

Expand the following using place values.

(a)23_{five} (b) 203_{five}

(c) 101_{two} (d) 432_{five}

(d) 104_{five} (e) 121_{three}

CHANGING NON-DECIMAL BASES TO BASE TEN (DECIMAL BASE)

Examples.

1. Convert 123five to base ten.

123five =

1	2	3
5^2	5^1	5^0

$(1 \times 5^2) + (2 \times 5^1) + (3 \times 5^0)$

$(2 \times 5 \times 5) + (2 \times 5) + (3 \times 1)$

$25 + 10 + 3$

38ten

2. Convert 342five to base ten

3	4	2	five
5^2	5^1	5^0	

$(3 \times 5^2) + (4 \times 5^1) + (2 \times 5^0)$

$(3 \times 5 \times 5) + (4 \times 5) + (2 \times 1)$

$75 + 20 + 2$

97_{ten}

3. Convert 102_{three} to base ten

1	0	2	three
3^2	3^1	3^0	

$(1 \times 3^2) + (0 \times 3^1) + (2 \times 3^0)$

$(1 \times 3 \times 3) + (0 \times 3) + (2 \times 1)$

$9 + 0 + 2$

11 ten

4. Convert 101 two to base ten

1	0	1
2^2	2^1	2^0

two

$(1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0)$

$(1 \times 2 \times 2) + (0 \times 2) + (1 \times 1)$

$4 + 0 + 1$

5 ten

Activity

1. Convert 23_{five} to base ten.
2. Convert 420_{five} to decimal base.
3. Convert 123_{five} to base ten.
4. Convert 403_{five} to base ten.
5. Convert the following bases to base ten.

(a) 304_{five} (b) 424_{five} (c) 102_{three} (d) 202_{three} (e) 111_{two}

CHANGING FROM BASE TEN TO OTHER BASE

Note:

Factorise using the required base.

Examples.

1. Change 7_{ten} to base five.

Base	No	Rem
5	7	2
5	1	1
	0	1

$7_{ten} = 12_{five}$

2. Convert 38ten to base five

Base	No	Rem
5	38	3
5	7	2
5	1	1
	0	

38 ten = 123_{five}

3. Convert 58ten to base five

Base	No	Rem
5	58	3
5	11	1
5	2	2
	0	

58ten = 213 five

4. Convert 20_{ten} to base three

Base	No	Rem
3	20	2
3	6	0
3	2	2
	0	

20 ten = 202 three

5. Convert 15ten to base two.

Base	No	Rem
2	15	1
2	7	1
2	3	1
2	1	1
	0	

15ten = 1111two

Activity

1. Convert 9ten to base five.
2. Convert the following to base five.

(a) 25ten (b) 30ten (c) 29ten

(d) 42 ten (e) 55ten

3. Convert 30ten to base three
4. Convert 20 ten to base two.

ADDITION OF BASE NUMBERS

Example I

Add: $1\ 2\ 3_{five}$

$+\quad 1\ 1_{five}$

$1\ 3\ 4_{five}$

Example II

Add: $\overset{1}{1}\ 4\ 2_{five}$

$+\quad 2\ 1_{five}$

$2\ 1\ 3_{five}$

$6 \div 5 = 1$ rem 1

Example III

Add : $1\ 4\ \overset{1}{2}\ 3_{five}$

$+\quad 2\ 1\ 2_{five}$

$1\ 1\ 4\ 0_{five}$

$5 \div 5 = 1$ rem 0

$6 \div 5 = 1$ rem 1

Example IV

Add: $1\ \overset{1}{0}\ 2_{three}$

$+\quad 1\ 2_{three}$

$1\ 2\ 1_{three}$

$4 \div 3 = 1$ rem 1

Example V

Add : $1\ 1\ 0\ 1_{two}$

$+\quad 1\ 1\ 1_{two}$

$1\ 1\ 0\ 0_{two}$

$2 \div 2 = 1$ rem 0

$2 \div 2 = 1$ rem 0

$3 \div 2 = 1$ rem 1

Example VI

Add: $1\ 0\ 2_{three}$

$+\ 1\ 1\ 1_{three}$

$2\ 1\ 0_{three}$

$3 \div 3 = 1$ rem 0

Activity

1. Add: $1\ 2\ 3_{five}$

$+\ 3\ 1_{five}$

2. Add: $2\ 4\ 3_{five}$

$+\ 2\ 4_{five}$

3. Add: $4\ 0\ 3_{five}$

$+\ 1\ 2_{five}$

4. Add: $4\ 4\ 0_{five}$

$+\ 4\ 3_{five}$

5. Add: $1\ 1\ 1_{two}$

$+\ 1\ 1\ 1_{two}$

6. Add: $2\ 0\ 1_{three}$

$+\ 1\ 0\ 2_{three}$

7. Add: $1\ 2\ 2_{three}$

$+\ 2\ 1\ 1_{three}$

SUBTRACTION OF BASES

Note: The one regrouped is worth the value of the base.

Example I

Subtract

$$\begin{array}{r} \textbf{3}\ \ \textbf{5} \\ 4\ \ 0\ \ 2_{\text{five}} \\ -\ 1\ \ 1_{\text{five}} \\ \hline 3\ \ 4\ \ 1_{\text{five}} \end{array}$$

Example II

Subtract:

$$\begin{array}{r} \textbf{1}\ \ \textbf{4}\ \ \textbf{5+} \\ 2\ \ 0\ \ 1_{\text{five}} \\ -\ 4\ \ 2_{\text{five}} \\ \hline 1\ \ 0\ \ 4_{\text{five}} \end{array}$$

Example III

Subtract

$$\begin{array}{r} \textbf{5}\ \ \textbf{5+0 (5 +)} \\ 3\ \ 1\ \ 2_{\text{five}} \\ -\ 1\ \ 2\ \ 3_{\text{five}} \\ \hline 1\ \ 3\ \ 4_{\text{five}} \end{array}$$

Example IV

Subtract

$$\begin{array}{r} \textbf{0}\ \ \textbf{5+} \\ 1\ \ 2\ \ 4_{\text{five}} \\ -\ 4\ \ 3_{\text{five}} \\ \hline 3\ \ 1_{\text{five}} \end{array}$$

Example V

Subtract:

$$\begin{array}{r} \textbf{0}\ \ \textbf{2} \\ 1\ \ 1\ \ 0_{\text{two}} \\ -\ 1\ \ 0\ \ 1_{\text{two}} \\ \hline 0\ \ 0\ \ 1_{\text{two}} \end{array}$$

Example VI

Subtract:

$$\begin{array}{r} {\scriptstyle 0\ \ 0\ \ 2} \\ 1\ 1\ 0\ 1_{\text{two}} \\ -\ \ 1\ \ 1\ \ 1_{\text{two}} \\ \hline 1\ \ 1\ 0_{\text{two}} \end{array}$$

Example VII

Subtract:

$$\begin{array}{r} {\scriptstyle 1\ \ 3} \\ 2\ \ 0\ \ 2_{\text{three}} \\ -1\ \ 2\ \ 1_{\text{three}} \\ \hline 1\ \ 1_{\text{three}} \end{array}$$

Activity

1. Subtract $1\ \ 2\ \ 3_{\text{five}} - 1\ \ 2_{\text{five}}$

2. Subtract $4\ \ 0\ \ 2_{\text{five}} - 1\ \ 2_{\text{five}}$

3. Subtract $4\ \ 1\ \ 3_{\text{five}} - 3\ 1\ 4_{\text{five}}$

4. Subtract $3\ \ 2\ \ 4_{\text{five}} - 1\ \ 4\ \ 2_{\text{five}}$

4. Subtract: $3\ \ 2\ \ 4_{\text{five}} - 1\ \ 4\ \ 2_{\text{five}}$

5. Subtract $4\ \ 0\ \ 1_{\text{five}} - 1\ \ 2\ \ 3_{\text{five}}$

6. Subtract $2\ \ 0\ \ 1_{\text{three}} - 1\ \ 2\ \ 1_{\text{three}}$

7. Subtract $1\ \ 2\ \ 0_{\text{three}} - 1\ \ 0\ \ 1_{\text{three}}$

8. Subtract $1\ \ 0\ \ 1_{\text{two}} - \ \ 1\ \ 1_{\text{two}}$

MULTIPLICATION OF BASES

Example I

1. Multiply: 2_{five} x 3

3 five	6 ÷ 5 = 1 rem 1
X 2 five	
1 1 five	

2. Multiply: $4\ 2\ 1_{five}$ x 2

4 2 1 five	2 x 1 = 2
X 2	8 ÷ 5 = 1 rem 3
1 3 4 2 five	

3. Multiply: 40 five x 3

$2\ 4\ 0_{five}$	12 ÷ 5 = 2 rem 2
X 3	
$2\ 2\ 0_{five}$	

4. **Example IV**

1 2 three x 2	4 ÷ 3 = 1 rem 1
$1\ 1\ 2_{three}$	3 ÷ 3 = 1 rem 0
X 2_{three}	
$1\ 0\ 1_{three}$	

Activity

1. Multiply: 4_{five} x 3
2. Work out 12_{five} x 4
3. Work out 21_{five} x 3
4. 121_{five} x 3
5. 321_{five} x 2
6. 113_{five} x 3
7. Multiply: 21_{three} x 2

FINITE SYSTEM

Is a way of expressing numbers as remainders.

Counting in finite system

Examples

1. Express 11 in finite 5

   Soln

   $\frac{11}{5} = 2$ r 1

   11 = 1 (finite 5)

2. Write 17 in finite 5

   Soln

   17 ÷ 5 = 3 rem 2

   17 = 2 (finite 5)

3. Change 9 to finite 7

   Soln

   9 ÷ 7 = 1 rem 2

   9 = 2 (finite 7)

4. Express 23 in finite 8.

   Soln

   23 ÷ 8 = 2 rem 7

   23 = 7 (finite 8)

5. Write 32 in finite 8

   Soln

   32 ÷ 4 = 4 rem o

   32 = 0 (finite 8)

Activity

1. Express the following in finite 5

(a) 9

(b) 5

(c) 13

(d) 19

(e) 20

2. Change the following to finite 7

(a) 22

(b) 33

(c) 8

(d) 53

(e) 49

3. Write the following in finite 8.

(a) 14

(b) 16

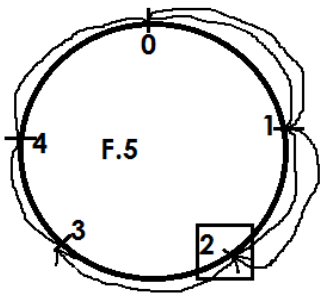
(c) 26

(d) 38

(e) 47

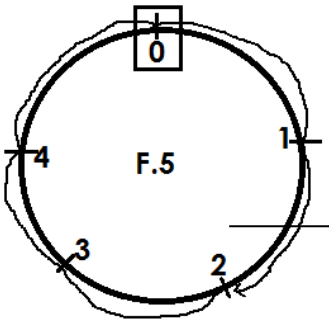
ADDITION IN FINITE USING ADAIL

1. Add: 3 + 4 = _____ (finite 5) using a dail.



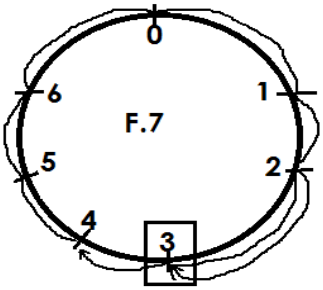
∴ 3 + 4 = 2 (finite 5)

2. Using a dail, add: 2 + 3 = _____(finite 5)



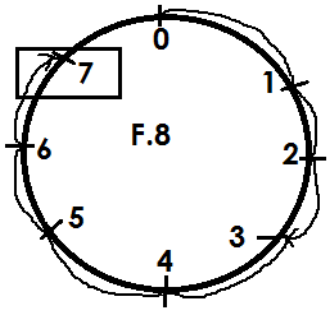
∴ 2 + 3 = 0 (finite 5)

3. _____ (finite 7)



∴ 4 + 6 = 3 (finite 7)

4. Using a dail, work out: 3 + 4 = _____ (finite 8)



∴ 3 + 4 = 7 (finite 8)

Activity

Using a dail, add the following.

a) 2 + 1 = _____ (finite 5)
b) 4 + 2 = _____ (finite 5)
c) 5 + 4 = _____ (finite 7)
d) 2 + 3 = _____ (finite 7)
e) 3 + 5 = _____ (finite 7)
f) 6 + 5 = _____ (finite 7)
g) 4 + 3 = _____ (finite 8)
h) 5 + 3 = _____ (finite 8)
i) 7 + 4 = _____ (finite 8)

ADDITION IN FINITE WITHOUT USING A DAIL

Note:

When the sum obtained is less than the given finite, maintain the sum as the answer without dividing by the finite.

Example I

Add: 4 + 3 = _____ (finite 5)

Soln

4 + 3 = _____ (finite 5)

7 ÷ 5 = 1 rem 2

Therefore 4 + 3 = 2 (finite 5)

Example II

Add : 2 + 4 =_____ (finite 7)

2 + 4 = 6 (finite 7)

Example

Work out: 2 + 1 + 6 = _____ (mod 7)

2 + 1 + 6 = _____ (mod 7)

$\frac{9}{7}$ = 1 rem 2

2 + 1 + 6 = 2 (mod 7)

Activity

1. **Add the following**

(a) 1 + 1 = _____ (finite 5)

(b) 2 + 3 = _____ (finite 5)

(c) 5 + 4 = _____ (finite 7)

(d) 3 + 4 = _____ (finite 5)

(e) 6 + 6 = _____ (finite 7)

(f) 4 + 2 = _____ (finite 5)

(g) 2 + 3 + 4 =_____ (finite 6)

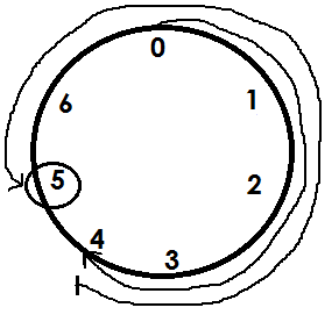
(h) 6 + 5 + 1 =_____ (finite 12)

(i) 4 + 3 + 5 = _____ (finite 8)

SUBTRACT INFINITE USING ADAIL

Example I

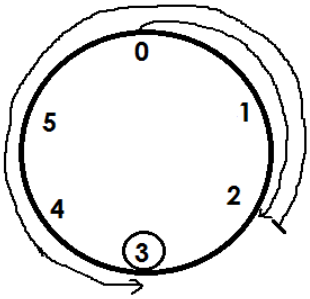
Work out: 4 – 6 = _____(mod 7)



∴ 4 - 6 = 5 (mod 7)

Example 2

Subtract: 2 – 4 = _____ (finite 5)



∴ 2 - 4 = 3 finite 5.

Activity

1. Workout the following using a dail.

(a) 2 - 4 = _____ (mod 6)

(b) 4 - 7 = _____ (finite 12)

(c) 3 – 5 = _____ (mod 7)

(d) 5 – 8 = _____ (mod 9)

(e) 1 - 4 = _____ (mod 4)

(f) 2 – 3 = _____ (mod 5)

(g) 1 - 4 = _____ (finite 6)

(h) 5 – 7 = _____ (finite 8)

PATTERNS AND SEQUENCES

TOPIC OUTLINE

- Types of numbers
- Multiples of numbers
- Prime factorization of numbers
- Finding LCM by prime factorization
- Finding LCM by listing multiples
- Application of LCM
- Factors of a number
- Finding the GCF by prime factorization
- Representing prime factors on venn diagrams
- Finding number which has been prime factorized
- Finding the missing prime factor
- Finding the un-known number given prime factors on a venn diagram
- Square roots of numbers
- Magic square

Review of P.4 Work on G.C.F and L.C.M from listed factors and multiples

Mk MTC Bk. 5 pg 80 – 82

TYPES OF NUMBERS.

1) Whole numbers ⟹ 0, 1, 2, 3, 4, 5, …

2) Counting numbers/ Natural numbers ⟹ 1, 2, 3, 4, 5,.

3) Odd numbers: These are numbers that give a remainder when divided by 2

⟹ {1, 3, 5, 7, …

4) Even numbers: These are ones that are exactly disable by two

⟹ {0, 2, 4, 6, 8, 10, …}

5) Prime numbers: These are numbers with only two factors.

⟹ {2, 3, 5, 7, 11, 13,17,19,23,29,31,37,41,43,47,53,59,61,67,71,73,79,83,89,97 …}

6) Composite numbers: These are numbers with more than 2 factors

⟹ {4, 6, 8, 9, 10, 12, …}

7) Square numbers: These are numbers got by multiplying a number by its self.

⟹ {1, 4, 9, 16, 25, …}

8) Triangular numbers: These are got by adding consecutive counting numbers.

{1, 3, 6, 10, 15, 21…}

$^{+}2$ +3 +4 +5 $^{+}6$

Guiding questions

1. Given that set R={even numbers less than 10}. List down the elements in set R.
2. If set T={prime numbers less than 20}. Find n(T)
3. Find the sum of the 4th and 5th triangular numbers
4. What is the 7th composite number?

5. Find the next number in the sequence:

a) 5,10,15,20,……., …….

b) 1,3,6,10,15,21, …….

c) 2,3,5,7,……., ……

d) 1,4,9,16,25,…….

e) Work out the square of 9

REFERENCE: Understanding MTC bk 6 pg 81 – 84

Mk Bk 5 Pg 80 – 89

Multiples of a number

A multiple is a product of a number and counting numbers.

Examples

1. List down the multiples of 6

x	1	2	3	4	5	6	7
6	6	12	18	24	30	36	42

M_6={6,12,18,24,30,36,42…}

2. List down the multiples of 7 between 20 and 50

x	1	2	3	4	5	6	7	8	9
7	7	14	21	28	35	42	49	56	63

M_7={21,28,35,42 and 49}

3. Find the first five multiples of 8

x	1	2	3	4	5
8	8	16	24	32	40

M_8={8,16,24,32 and 40}

Activity

1. List down the multiples of the following

a) 3

b) 9

c) 12

2. Find the first six multiples of 4
3. List down all the multiples of 5 between 16 and 32
4. List down all the multiples of 13 less than 60

PRIME FACTORIZATION OF NUMBERS

Prime factorizing means dividing a number by its prime factors.

We use prime factors when prime factorizing e.g. = {2, 3, 5, 7, 11, 13 …}

Example I

Prime factorize 18.

We can either use a ladder or a factor tree. i.e.

2	18
3	9
3	3
	1

We can represent the prime factors as follows.

Set notation / subscript form

$18 = \{2_1, 3_1, 3_2\}$

Multiplication form

$18 = 2 \times 3 \times 3$

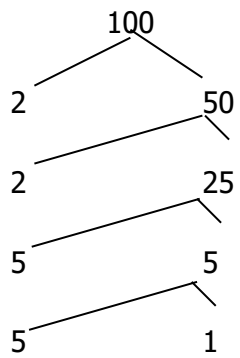
power form (expanded form)

$18 = 2^1 \times 3^2$

Example II

Prime factorize 100.

* Take note of factors of 100 that are prime numbers.



Subscript form ÷

$100 = \{2_1, 2_2, 5_1, 5_2\}$

Multiplication form

$100 = 2 \times 2 \times 5 \times 5$

Power form

$100 = \underline{2^2 \times 5^2}$

Ref: MK MTC BK. 5 pg 84 – 85, Fountain MTC BK. 5 pg 106 – 108

Understanding MTC Bk. 5 Pg. 79 – 81

FINDING L.C.M BY PRIME FACTORIZING

Example

Find the L.C.M of 4 and 12 by prime factorization.

2	4	12
2	2	6
3	1	3
	1	1

L.C.M = 2 x 2 x 3

= 4 x 3

= **12**

Example II

Find the L.C.M of 12 and 20.

2	12	20
2	6	10
3	3	5
5	1	5
	1	1

L.C.M = 2 x 2 x 3 x 5

= 4 x 15

= 60

REFERENCE: MK MTC Bk 5 pg 86.

Finding the Lowest Common Multiple (LCM) of numbers by listing

Examples

1. Work out the LCM of 6 and 4

   M_6={6,12,18,24,30,36,…}

   M_4={4,8,12,16,20,24,28,…}

   CM={12,24,…}

   LCM=12

2. Work out the LCM of 12 and 10

   M_{12}={12,24,36,48,60,72,…}

   M_{10}={10,20,30,40,50,60,70,…}

   CM={60,…}

   LCM=60

   Activity

1. Use the method of listing multiples tom work out the LCM of :

a) 2 and 3

b) 4 and 5

c) 7 and 8

d) 5 and 7

e) 12 and 1

Application of LCM

Examples

1. Find the least number of mangoes that can be shared by 5 or 6 children leaving no remainder.

2	5	6
3	5	3
5	5	1
	1	1

2x3x5=30 mangoes

2. Work out the least number of books which can be exactly shared by either 6 or 8 children leaving no remainder.

2	6	8
2	3	4
2	3	2
3	3	1
	1	1

2x2x2x3=24

24 books

3. Work out the least number of sweets which can be shared by 4 or 5 children leaving a remainder of 3

2	4	5
2	2	5
5	1	5
	1	1

LCM + Remainder

2x2x5 +3

20+3

23 sweets

4. What is the lowest number when divided by either 4 or 6 leaves a remainder of 2.

2	4	6
2	2	3
3	1	3
	1	1

LCM + Remainder

2x2x3 + 2

12+ 2

14

Activity

1. Work out the least number which is exactly divisible by 5 and 8

2. Work out the least number when divided by 6 and 10 leaves a remainder of 4

3. Work out the least number of books which can be shared by 12 0r 10 children leaving no remainder.

4. Find the least number of apples which can be shared by 8 or 12 children leaving no remainder.

5. What is the least number of pencils which can be shared by 6 or 9 children leaving a remainder of 3 pencils.

Factors of a number

A factor is a number which exactly divides a given number

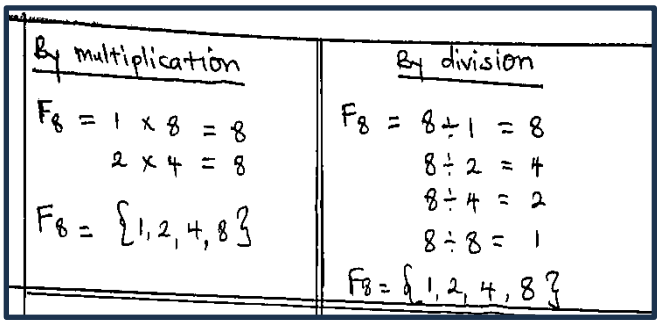
They are also called **divisors**

Factors can be got in two ways i.e

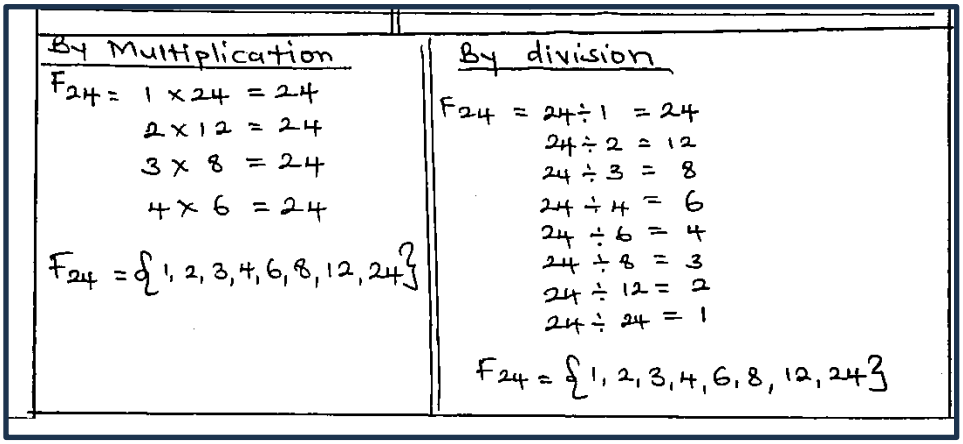
a) by division
b) by multiplication

Examples

1. List down the factors of 8



2. List down all the factors of 24



Activity

List down all the factors/divisors of the following

a) 10
b) 12
c) 30
d) 20
e) 36

2. Find the greatest number which can divide 18 and 24 leaving no remainder
3. Work out the largest number which can divide 20 and 12 leaving no remainder

FINDING G.C.F BY PRIME FACTORIZING

Example:

Find the G.C.F of 6 and 8

2	6	8
	3	4

G.C.F = 2

Example II

Find the G.C.F of 24 and 36.

2	24	36
2	12	18
3	6	9
	2	3

G.C.F = 2 x 2 x 3

= 4 x 3 = **12**

Reference: NEW MK. BK 5 PAGE 87

Activity

work out the GCF of the following

a) 12 and 18
b) 10 and 12
c) 20 and 24
d) 30 and 36
e) 24 and 32

REPRESENTING PRIME FACTORS ON VENN DIAGRAMS

Example I

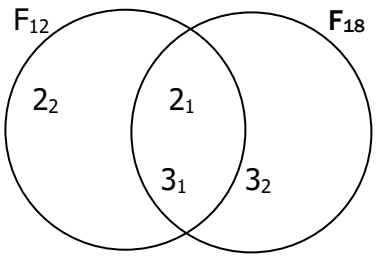
Represent the prime factors of 12 and 18 on a Venn diagram.

2	12
2	6
3	3
	1

2	18
3	9
3	3
	1

$F_{12} = \{ 2_1, 2_2, 3_1 \}$

$F_{18} = \{ (2_1), (3_1), (3_2) \}$



a) Find the G>C>F of 12 and 18.

G.C.F product of the intersection

$F_{12} \cap F_{18} = \{2_1, 3_1\}$

$\therefore$ **G.C.F = 2 x 3**

<u>6</u>

L.C.M = product of the union.

$F_{12} \cup F_{18} = \{2_1, 2_2, 3_1, 3_2\}$

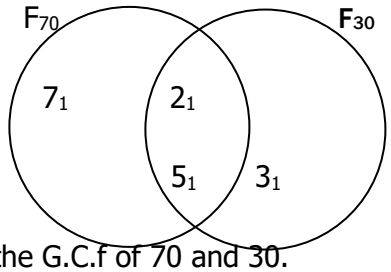
= 2 x 2 x 3 x 3

= 4 x 9

= **<u>36</u>**

<u>Example 2</u>

Below is a Venn diagram showing factors.



a) Find the G.C.f of 70 and 30.

G.C.F = product of the intersection

$F_{70} \cap F_{30} = \{2_1, 5_1\}$

G.C.F = 2 x 5

= **<u>10</u>**

b) Find the L.C.M of 20 and 70.

L.C.M =product of the union

L.C.m = 2_1 x 3_1 x 5_1 x 7_1

= 6 x 35

L.C.M = **<u>210.</u>**

Activity:

1. Prime factorise the following and give your answer in set notation or subscript form

a) 24

b) 36

c) 30

2. Express the following numbers as a product of their prime factors

a) 20

b) 48

c) 60

Fountain MTC Bk. 5 Pg 118 - 119

MK MTC BK. 6 pg. 86

Finding number which has been factorized

Examples

1. Find the number which has been factorized to get ;

   $\{2_1,2_2,3_1,3_2\}$

   2x2x3x3

   36

2. If $PF_Y=\{2_1,3_1,5_1\}$. Find the value of y

   y= 2x3x5

   y= 30

3. If $W=2^2X3^1$. Find the value of w

   w= 2x2x3

   w= 12

Activity

1. Find the number which has been factorized to get :

a) $\{2_1,2_2,3_1,5_1\}$

b) $\{3_1,3_2,5_1\}$

c) $\{2_1,3_1,3_2\}$

d) 2^3x3^1

e) 2^2x5^2

2. If $PF_r=\{2_1,3_1,3_2,5_1\}$, find the value of r

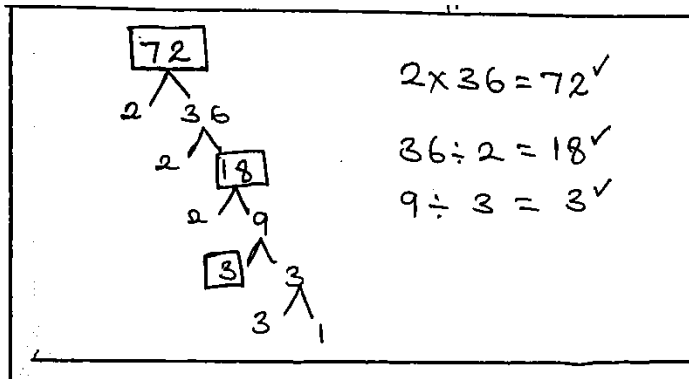
3. Workout the value of q if $PF_q=\{3_1,5_1,5_2\}$

Finding the missing prime factor

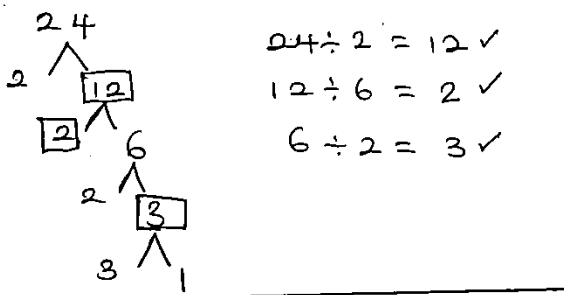
Examples

1. Complete the factor tree below

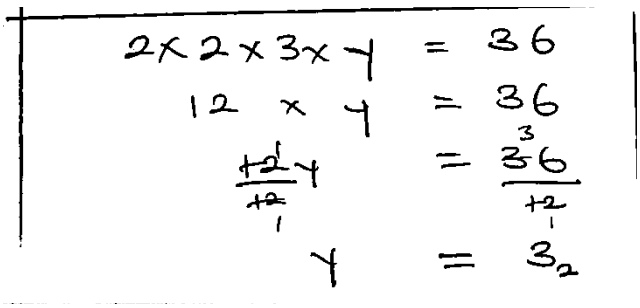
a) 72



b) 24



2. Given that $PF_{36}=\{2_1,2_2,3_1,Y\}$. Find the value of y.



3. Work out the value of r if $PF_{20}=\{2_1,2_2,r\}$

$2 \times 2 \times r = 20$

$4 \times r = 20$

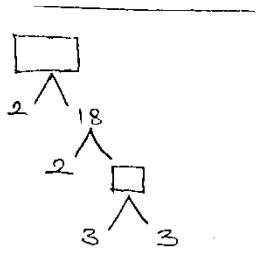
$\frac{4r}{4} = \frac{20}{4}$

$r = 5_1$

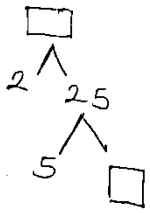
Activity

1. Complete the factor tree below

a)

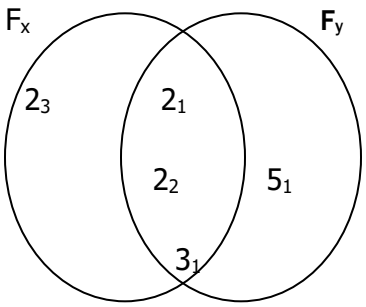


b)



2. Given that $pf_{24}= \{2_1,2_2,2_3,w\}$. Find the value of w
3. If $PF_{30}=\{2_1,y,5_1\}$.Find the value of y
4. If $PF_{16}=\{2_1,2_2,2_3,r\}$. Work out the value of r.

FINDING THE UNKNOWN NUMBER GIVEN PRIME FACTORS ON A VENN DIAGRAM



a) Find the value of x.

$F_x = \{2_1, x\ 2_2, x\ 3_1\}$

$x = 4 \times 6$

$x =$ **24**

b) Find the value of y.

$F_y = \{2_1, 2_2, 3_1, 5_1\}$

$y = 2_1 \times 2_2 \times 3_1 \times 5_1$

$y = 2 \times 2 \times 3 \times 5$

$y = 4 \times 15$

$y =$ **60**

c) Work out the LCM of x and y

LCM= Product of all prime factors in the union

$\{2_1, 2_2, 2_3, 3_1, 5_1\}$

2x2x2x3x5

120

d) Work out the GCF of x and y

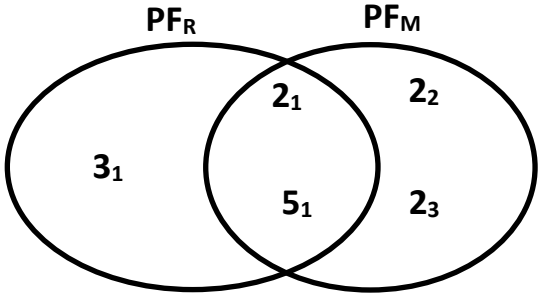
GCF= Product of the intersection

2x2x3

12

Activity

1. Use the venn diagram below to answer the questions that follow



a) Find the value of ;

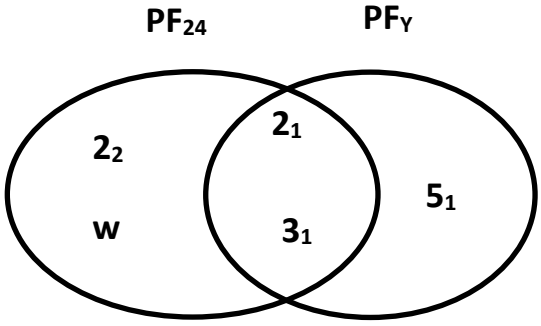
i. R

ii. M

b) Work out the LCM of PF_R and PF_M

c) Work out the GCF of PF_R and PF_M

2. Given the venn diagram below, use it to answer the questions that follow



a) Find the value of

i. y

ii. w

b) Work out the GCF of PF_{24} and PF_Y

c) Work out the LCM of PF_{24} and PF_Y

REFERENCE: MK BK 6 Pg 88 – 89

SQUARE ROOTS OF NUMBERS

Review of square numbers.

i.e. {1, 4, 9, 16, 25, 36, 49, 64, 81, …}

A square number is got by multiplying a number by itself.

A square root is a number that is multiplied by its self to give a square number.

The symbol for square root is $\sqrt{}$

Example

Find the square root of 36.

2	36
2	18
3	9
3	3
	1

$\sqrt{36} = \sqrt{(2\times 2)\times(3\times 3)}$

$\sqrt{36}$ = 2 x 3

= **6**

Example:

Find the square root of 100.

2	100
2	50
5	25
5	5
	1

$\sqrt{100} = \sqrt{(2x2)x(5x5)}$

= 2 x 5

= **10**

Activity

Work out the square root of

a) 25
b) 9
c) 64
d) 49
e) 144

Ref: MK MTC BK. 5 pg. 89, MK MTC BK. 6 pg. 95

Magic square

A magic square is a square whose sum of digits vertically, horizontally and diagonally is the same.

NOTE:

The sum of digits vertically, horizontally or diagonally is referred to as a magic sum.

Example

Given the magic sum below

8	1	x
w	5	y
4	p	2

Find the magic sum

8+5+2

15

b) Find the value of ;

i. x

x + 8 +1 = 15
x +9 =15
x + 9-9 =15-9
x = 6

ii. y

y + x + 2 = 15
y + 6 + 2 =15
y +8 = 15
y + 8-8 =15 -8
y = 7

(iii) p + 1 +5 = 15

p + 6 =15

p +6-6 = 15 -6

p = 9

(iv) w + 8 + 4 =15

w + 12 = 15

w +12-12 = 15-12

w = 3

Activity

1. **Given the magic square below**

4	**d**	**8**
b	**10**	**c**
a	**2**	**16**

(a) Work out the magic sum

(b) Find the value of ;

i. **a**
ii. **b**
iii. **c**
iv. **d**

2. Complete the magic squares below

5	**x**	**y**
w	**8**	**p**
9	**4**	**11**

---	**13**	**---**
---	**9**	**----**
7	**5**	**15**

b	**c**	**8**
d	**7**	**9**
6	**11**	**a**

FRACTIONS (FIRST PART)

A fraction is part of a whole.

REVIEW

- ✓ Types of fractions
- ✓ Expressing improper fractions as mixed numbers.
- ✓ Expressing mixed numbers as improper fractions.
- ✓ Equivalent fractions.

REDUCING FRACTIONS

Fractions can be reduced using prime factors.

Example 1

Reduce $\frac{12}{24}$

F_{12} =

2	12
2	6
3	3
	1

F_{24} =

2	24
2	12
2	6
3	3
	1

$$\frac{12}{24} = \frac{2 \times 2 \times 3}{2 \times 2 \times 2 \times 3}$$

$$= \frac{1}{2}$$

Example II

Reduce $\frac{75}{100}$ to its lowest terms.

F_{12} =

3	75
5	25
5	5
	1

F_{100} =

2	100
2	50
5	25
5	5
	1

$$\frac{75}{100} = \frac{3 \times 5 \times 5}{2 \times 2 \times 5 \times 5}$$

$= \frac{3}{4}$

Example III

Reduce $\frac{18}{24}$ to its lowest terms.

Note:

Fractions can also be reduced by dividing both numerator and denominator by GCF.

$\frac{18}{24} = \frac{18 \div 6}{24 \div 6}$ $F_{18} = \{1, 2, 3, 6, 18\}$

$F_{24} = \{1, 2, 3, 4, 6, 8, 12, 24\}$

$= \frac{3}{4}$

Activity

Reduce the following fractions to their lowest forms.

1) $\frac{2}{4}$ 2) $\frac{8}{12}$ 3) $\frac{6}{18}$ 4) $\frac{30}{90}$ 5) $\frac{32}{56}$

6) $\frac{24}{72}$ 7) $\frac{21}{42}$ 8) $\frac{18}{36}$

ORDERING FRACTIONS

Note:

- Ascending order means from smallest to biggest.
- Descending order means from biggest to smallest.
- Find the LCM of denominators and the multiply it by all fractions to get their value in whole form.

Example I

Arrange the following fractions in ascending order

$\frac{1}{3}, \frac{1}{2}, \frac{1}{4}$

Soln

LCM =

2	3	2	4
2	3	1	2
3	3	1	1
	1	1	1

$= 2 \times 2 \times 3$

$= 12$

$\frac{1}{3}$ 4 $\frac{1}{3} \times 12$ 4	$\frac{1}{2}$ 5 $\frac{1}{2} \times 10$ 5	$\frac{1}{4}$ 3 $\frac{1}{4} \times 12$ 3

Ascending order = $\frac{1}{4}, \frac{1}{3}, \frac{1}{2}$

Example II

Arrange the following fraction in a descending order: $\frac{5}{8}, \frac{7}{12}, \frac{3}{8}$

LCM =

2	8	12	8
2	4	6	4
2	2	3	2
3	1	3	1
	1	1	1

LCM = 2 x 2 x 2 x 3
= 24

$\frac{5}{8}$ 3 $\frac{5}{8} \times 24$ 5 x 3 15	$\frac{7}{12}$ 2 $\frac{7}{12} \times 24$ 2 x 7 14	$\frac{3}{8}$ 3 $\frac{3}{8} \times 24$ 3 x 3 9

Descending order = $\frac{5}{8}, \frac{7}{12}, \frac{3}{8}$

Activity

1. Arrange the following fractions in ascending order.

(a) $\frac{3}{4}$, $\frac{2}{3}$, $\frac{1}{2}$

(b) $\frac{5}{6}$, $\frac{5}{8}$, $\frac{5}{12}$

(c) $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{6}$

(d) $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{9}$

2) Arrange the following fractions in a desending order

(a) $\frac{2}{3}$, $\frac{1}{12}$, $\frac{5}{6}$

(b) $\frac{7}{10}$, $\frac{5}{6}$, $\frac{12}{18}$

(c) $\frac{2}{3}$, $\frac{5}{9}$, $\frac{3}{5}$

(d) $\frac{1}{4}$, $\frac{2}{5}$, $\frac{3}{8}$

ADDITION OF SIMPLE FRACTIONS

Note:

All answers must be in their simplest forms i.e., either a simplifies proper fraction or a mixed number.

Examples

Add: $\frac{1}{4} + \frac{1}{2}$

LCM of the denominators =

2	4	2
2	2	1
	1	1

2 x 2 = 4

$\frac{1}{4} + \frac{1}{2} \frac{(\frac{1}{4} \times 12) + (\frac{1}{2} \times 4)}{12}$

Or

$= \left(\frac{1 \times 2}{4 \times 2}\right) + \left(\frac{1 \times 4}{2 \times 4}\right)$

$= \frac{2}{8} + \frac{4}{8}$

$= \frac{(1 \times 3) + (1 \times 2)}{12}$

$= \frac{1 + 2}{4} = \frac{3}{4}$

$= \frac{\cancel{6}\ 3}{\cancel{8}\ 4}$

$= \frac{3}{4}$

Example II

Work out: $\frac{5}{6} + \frac{3}{8}$

$\frac{5}{6} + \frac{3}{8} = \left[\frac{5}{\cancel{6}_1} \times \cancel{24}^{4}\right] + \left[\frac{3}{\cancel{8}_1} \times 24^{3}\right]$

$= \frac{(5 \times 4) + (3 \times 3)}{24}$

$= \frac{20 + 9}{24}$

$= \frac{29}{24}$ 1 rem 5

$= 1\frac{5}{24}$

LCM =

2 | 6	8	
2 | 3	4	
2 | 3	2	
3 | 3	1	
1 | 1	1	

2 x 2 x 2 x 3

= 24

Activity

Add the following fraction.

1. $\frac{1}{3} + \frac{1}{2}$
2. $\frac{2}{3} + \frac{1}{5}$
3. $\frac{1}{5} + \frac{1}{3}$
4. $\frac{2}{5} + \frac{1}{6}$
5. $\frac{2}{9} + \frac{1}{6}$
6. $\frac{7}{15} + \frac{1}{6}$

7. $\frac{7}{8} + \frac{2}{3}$

8. $\frac{3}{4} + \frac{1}{5}$

ADDITION OF FRACTIONS TO WHOLE NUMBERS

Examples

Workout

$4\frac{1}{6} + 10$

$4\frac{1}{6} + 10 = (4 + 10) + \frac{1}{16}$

$= 14 + \frac{1}{6}$

$= 14\frac{1}{6}$

Example III

Add $4\frac{3}{8} + \frac{1}{2}$

Soln

$4 + (\frac{3}{8} + \frac{1}{2})$

4

$4 + (\frac{3}{8} + 8) + (\frac{1}{2} \times 8)$

$4 + (3 + 4)$

8

$4 + \frac{7}{8}$ $4\frac{7}{8}$

Activity

Workout the following fractions

1. $\frac{1}{2} + 8$
2. $4\frac{3}{8} + 9$
3. $5 + 4\frac{2}{3}$
4. $\frac{5}{7} + 9$
5. $10 + 2\frac{3}{4}$
6. $15 + 5\frac{3}{9}$
6. $15 + 5\frac{3}{7}$
7. $1\frac{1}{2} + \frac{1}{4}$
8. $6\frac{1}{7} + \frac{3}{7}$
9. $1\frac{7}{8} + \frac{1}{2}$

ADDITION OF MIXED NUMBERS

Note:

Change the mixed numbers to improper fraction.

Use the LCM approach to work out

Example I

Work out: $1\frac{1}{2} \times 3\frac{1}{4}$

Soln

$1\frac{1}{2} + 3\frac{1}{4} = \frac{3}{2} + \frac{13}{4}$

4

2

$= \frac{(\frac{3}{2} \times 4) + (\frac{13}{4} \times 4)}{4}$

$= \frac{(3 \times 2) + (13 \times 1)}{4}$

$= \frac{6 + 13}{4}$

$= 4\frac{3}{4}$

Work out : $1\frac{1}{2} + 1\frac{1}{4} + 1\frac{1}{3}$

LCM = 12

$1\frac{1}{2} + 1\frac{1}{4} + 1\frac{1}{3} = \frac{3}{5} + \frac{5}{4} + \frac{4}{3}$

6 3 4

$= \frac{(\frac{3}{2} \times 12) + (\frac{5}{4} \times 12) + (\frac{4}{5} \times 12)}{12}$

$= \frac{(3 \times 6) + (5 \times 3) + (4 \times 4)}{12}$

$= \frac{18 + 15 + 16}{12}$

$= 49$ 4 rem 1

$= 4\frac{1}{12}$

Work out : $\frac{1}{15} + 1\frac{1}{3} + \frac{3}{5}$

$= \frac{1}{15} + 1\frac{1}{3} + \frac{3}{5} = \frac{1}{15} + \frac{4}{3} + \frac{3}{5}$

5 3

$= \frac{(\frac{1}{15} \times 15) + (\frac{4}{3} \times 15) + (\frac{3}{5} \times 15)}{15}$

$= \frac{1 + 20 + 9}{15}$

$= \frac{30}{15}2$

$= 2$

Activity

Workout the following

1. $3\frac{1}{2} + 4\frac{1}{3}$
2. $4\frac{1}{5} + 2\frac{1}{2}$
3. $1\frac{1}{6} + \frac{7}{15} + 1\frac{1}{3}$
4. $1\frac{2}{5} + \frac{2}{10} + \frac{3}{5}$
5. $\frac{1}{3} + 5\frac{1}{4} + \frac{1}{6}$
6. $\frac{3}{4} + 4\frac{1}{8} + 2\frac{5}{8}$
7. $2\frac{4}{9} + 3\frac{2}{9} + 3\frac{1}{2}$
8. $2\frac{1}{15} + 1\frac{3}{5} + 1\frac{3}{5}$
9. $3\frac{1}{6} + 2\frac{1}{5} + \frac{1}{15}$
10. $\frac{3}{5} + 1\frac{3}{4} + 4\frac{1}{2}$

WORD PROBLEMS INVOLVING ADDITION OF FRACTIONS

Example I

Tom read $\frac{1}{2}$ of the book on Monday and $\frac{1}{5}$ on Tuesday. Find the fraction of the book he read altogether.

Soln

$$\frac{1}{2} + \frac{1}{5} = \frac{(\frac{1}{2} \times 10) + (\frac{1}{5} \times 10)}{10}$$

$$= \frac{(1 \times 5) + (1 \times 2)}{10}$$

$$= \frac{5 + 2}{10}$$

$$= \frac{7}{10}$$

Example II

A teacher spent $\frac{1}{3}$ an hour giving examples, $1\frac{1}{2}$ for exercise and $1\frac{1}{4}$ for making. How long was the lesson?

$\frac{1}{3} + 1\frac{1}{2} + \frac{1}{4}$

4 6 3

$=(\frac{1}{3} \times 12) + (\frac{3}{2} \times 12) + (\frac{1}{4} \times 12)$

$=(1 \times 4) + (3 \times 16) + (1 \times 3)$

$= 4 + 18 + 3$

12

$= \frac{25}{12} = 2$ rem 1

$= 2\frac{1}{12}$ hrs

Activity

1. Mark filled $\frac{1}{2}$ of a tank with water in the morning and $\frac{2}{5}$ in the afternoon. Find the fraction of the tank that he filled with water.
2. James covered $\frac{1}{6}$ of the journey by bus and $\frac{4}{9}$ by taxi. What fraction of the journey did he cover?
3. Oluku gave $3\frac{1}{9}$ cakes to the son and $\frac{4}{9}$ to the daughter. How many cakes did he give out?
4. Mary had $1\frac{1}{2}$ sweets, Jane had $2\frac{3}{4}$ sweets and Rose had $\frac{3}{4}$ of a sweet. How many sweets did they have altogether?
5. $\frac{5}{18}$ of the wire is painted red, $\frac{1}{18}$ white and $\frac{1}{9}$ blue. How long is the wire?
6. $\frac{2}{3}$ of the land is for grazing, $\frac{1}{12}$ for planting crops and $\frac{1}{8}$ for construction. Find the size of the whole land.

SUBTRACTION OF SIMPLE FRACTIONS

Examples

Simplify: $\frac{3}{8} - \frac{1}{8}$

Son

$$\frac{3}{8} - \frac{1}{8} = \frac{3 - 1}{8}$$

$$= \frac{2}{8} \quad \frac{1}{4}$$

$$= \frac{1}{4}$$

Workout: $\frac{1}{2} - \frac{1}{3}$ **LCM**= 6

$$\frac{1}{2} - \frac{1}{3} = \frac{(\frac{1}{12} \times 6) - (\frac{1}{3} \times 6)}{6}$$

$$= \frac{(1 \times 3) - (1 \times 2)}{6}$$

$$= \frac{1}{6}$$

Simplify $2\frac{1}{5} - 1\frac{1}{2}$ LCM = 10

Soln

$$2\frac{1}{5} - 1\frac{1}{2} = \frac{11}{5} - \frac{3}{2}$$

$$= \frac{(\frac{11}{5} \times 10) - (\frac{3}{2} \times 10)}{10}$$

$$= (\frac{11}{5} \times 10) - (\frac{11}{2} \times 10)$$

$$= \frac{(11 \times 2) - (3 \times 5)}{10}$$

$$= \frac{22 - 15}{10}$$

$$= \frac{7}{10}$$

Example IV

Work out: $5 - 1\frac{1}{3}$

Soln LCM = 3

$5 - 1\frac{1}{3} = \frac{5}{1} - \frac{4}{3}$

$= \frac{(5 \times 3) - (4 \times 3)}{3}$

$= \frac{(5 \times 3) - (4 \times 1)}{3}$

$= \frac{15 - 4}{3}$

$= \frac{11}{3}$ 3 rem 2

$= 3\frac{2}{3}$

Activity

Work out the following

(1) $\frac{4}{9} - \frac{1}{9}$

(2) $\frac{5}{6} - \frac{1}{6}$

(3) $\frac{3}{4} - \frac{1}{4}$

(4) $\frac{5}{7} - \frac{2}{3}$

(5) $\frac{2}{3} - \frac{1}{4}$

(6) $\frac{3}{4} - \frac{1}{3}$

(7) $2\frac{1}{5} - \frac{1}{10}$

(8) $3\frac{3}{4} - 1\frac{1}{2}$

(9) $3\frac{3}{4} - 1\frac{1}{2}$

(10) $6 - 1\frac{1}{9}$

(11) $8 - 3\frac{1}{4}$

(12) $12 - 5\frac{1}{6}$

WORD PROBLEMS INVOLVING FRACTIONS

Examples

Matama had $\frac{1}{6}$, what fraction of the sugar cane remained?

Soln LCM

Remaining fraction = $\frac{4}{5} - \frac{1}{6}$

$$= \frac{\left(\frac{4}{5} \times 30\right) - \left(\frac{1}{6} \times 30\right)}{30}$$

$$= \frac{24 - 5}{30}$$

$$= \frac{19}{30}$$

Example II

James had $5\frac{1}{3}$ cakes, he gave out $2\frac{1}{2}$ to mark. How many cakes did he remain with?

Soln

Note:

- Change mixed to improper

Remaining cakes = $5\frac{1}{3} - 2\frac{1}{2}$

$$= \frac{16}{3} - \frac{5}{2}$$

$$= \frac{\left(\frac{16}{3} \times 6^{2}\right) - \left(\frac{5}{2} \times 6^{3}\right)}{6}$$

$$= \frac{(16 \times 2) - (5 \times 3)}{6}$$

$$= \frac{32 - 15}{6}$$

$$= \frac{17}{6} \quad 2 \text{ rem } 5$$

$$= 2\frac{5}{6}$$

Example III

My father had 6 plots of land. If he gave out $2\frac{1}{4}$ to the children, find the number of plots of land that he remained with.

Soln LCM 4

Remaining plots

$= 6 - 2\frac{1}{4}$

$= \frac{6}{1} - \frac{9}{4}$

$= \frac{\left(\frac{6}{1} \times 4\right) - \left(\frac{9}{\cancel{4}_1} \times \cancel{4}\right)}{4}$

$= \frac{24 - 9}{4}$

$= \cancel{15}\ 3 \text{ rem } 2$
$\cancel{4}$

$= 3\ \frac{3}{4}$ plots

Activity

1. A girl had a $\frac{1}{2}$ glass full of water and used $\frac{1}{3}$ to take medicine. What fraction of water remained in the glass?
2. Moses was given $\frac{3}{4}$ of a sugar cane and he also gave out $\frac{1}{6}$ to his friend. What fraction for the sugar cane did he remain wit?
3. Mukasa had $3\frac{1}{3}$ oranges. If he gave out $1\frac{3}{4}$ to Peter, find the fraction of the oranges that remained.
4. There were $16\frac{1}{3}$ boxes of books in the store, if $8\frac{1}{2}$ boxes were given to primary five East class, how many boxes remained in the store?
5. Okello bough 4kg of rice and $2\frac{1}{3}$ were used. How many kg of rice remained?
6. After covering $\frac{1}{9}$ of the journey by a taxi, Wandera walked the remaining distance. Find fraction of the remaining distance.
7. Out of the 8 bars of soap that were bought we have so far used $6\frac{1}{9}$. Find the remaining band of soap.

ADDITION AND SUBTRACTION OF FRACTIONS.

Note:

Re-arrange the given fraction starting with addition.

Example I

Work out: $\frac{1}{4} - \frac{1}{2} + \frac{1}{3}$

Soln LCM = 12

$$\frac{1}{4} - \frac{1}{2} + \frac{1}{3} = \frac{1}{4} + \frac{1}{3} - \frac{1}{2}$$

$$\frac{\left(\frac{1}{4} \times 12\right) + \left(\frac{1}{3} \times 12\right) - \left(\frac{1}{2} \times 12\right)}{12}$$

$$= \frac{(1 \times 3) + (1 \times 4)}{12} - 6$$

$$= \frac{(3 + 4) - 6}{12}$$

$$= \frac{7 - 6}{12} \quad \Big| \quad = \frac{1}{12}$$

Example II

Simplify: $\frac{1}{4} - 2\frac{1}{2} + 4\frac{1}{3}$

Soln LCM = 12

$$\frac{1}{4} - 2\frac{1}{2} + 4\frac{1}{3} = \frac{1}{3} - \frac{5}{2} + \frac{13}{3}$$

$$= \frac{1}{4} + \frac{13}{3} - \frac{5}{2}$$

$$= \left(\frac{1}{4} \times 12^{3}\right) + \left(\frac{13}{3} \times 12^{4}\right) - \left(\frac{5}{2} \times 12^{6}\right)$$

$$= \frac{(3 + 52) - 30}{12}$$

$$= \frac{55 - 30}{12}$$

$$= \frac{22}{12} \text{ 1 rem 10}$$

$$= 1\frac{10}{12}$$

$$= \mathbf{1\frac{5}{6}}$$

ACTIVITY

Workout the following

1. $\frac{2}{3} - \frac{5}{6} + \frac{3}{4}$

2. $\frac{90}{20} - \frac{4}{3} + \frac{7}{10}$

3. $\frac{1}{2} + \frac{4}{5} - \frac{7}{10}$

4. $\frac{2}{3} + \frac{3}{5} - \frac{7}{15}$

5. $\frac{5}{12} + \frac{7}{12} - \frac{11}{12}$

6. $\frac{5}{12} - 1\frac{1}{2} + 7\frac{1}{4}$

7. $2\frac{1}{6} + 3\frac{1}{2} + 2\frac{1}{3}$

8. $\frac{1}{4} - 4\frac{1}{3} + 6\frac{1}{2}$

MULTIPLICATION OF WHOLES AND FRACTIONS

Examples

Simplify: $\frac{1}{2}$ x 12

Soln

$\frac{1}{2} \times 12 = \frac{1 \times 12^{6}}{2_{1}}$

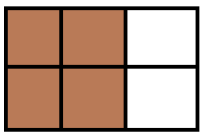
$= 1 \times 6$

$= 6$

Examples

2. Shade $\frac{2}{3}$ of the figure below

Soln



$= \frac{2}{3} \times 6$ parts

$= 4$ parts

Example II

Simplify: $\frac{1}{3} \times 10$

Soln

$\frac{1}{3} \times 10 = \frac{1 \times 10}{3}$

$= \frac{10}{3}$ 3 rem 1

$= 3\frac{1}{3}$

Examples III

Work out: $\frac{2}{3} \times 24$

Soln

$\frac{2}{3} \times 24 = \frac{2}{\cancel{3}} \times \cancel{24}\ 8$

$= 2 \times 8$

$= 16$

Example IV

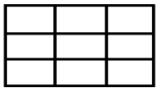
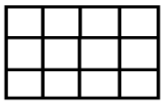
Simplify: $\frac{2}{7} \times 6$

$\frac{2}{7} \times 6 = \frac{2 \times 6}{7}$

$= \frac{12}{7}$ 1 rem 5

Activity

Simplify the following

1. $\frac{1}{3} \times 3$
2. $\frac{2}{3} \times 15$
3. $6 \times \frac{2}{3}$
4. $\frac{1}{4} \times 12$
5. $\frac{5}{6} \times 36$
6. $\frac{4}{9} \times 18$
7. $\frac{1}{3} \times 8$
8. $\frac{3}{4} \times 9$
9. $\frac{2}{5} \times 9$
10. Shade $\frac{2}{9}$ of 
11. Shade $\frac{3}{4}$ 

MULTIPLICATION OF FRACTIONS BY FRACTIONS

Note:

Multiply numerator by numerator and denominator by denominator.

Examples.

Simplify: $\frac{1}{3} \times \frac{1}{4}$

$\frac{1}{3} \times \frac{1}{4}$

$\frac{1 \times 1}{3 \times 4}$

$\frac{1}{12}$

Example II

Simplify: $\frac{2}{3} \times \frac{5}{6}$

$\frac{2}{6} \times \frac{5}{3} = \frac{\cancel{2}}{\cancel{6}} \times \frac{5}{3}$

$= \frac{5}{9}$

Activity

Simplify the following

1. $\frac{1}{2} \times \frac{1}{2}$

5. $\frac{2}{3} \times \frac{3}{4}$

2. $\frac{1}{2} \times \frac{1}{3}$

6. $\frac{3}{4} \times \frac{1}{3}$

3. $\frac{1}{8} \times \frac{1}{5}$

7. $\frac{3}{8} \times \frac{4}{5}$

4. $\frac{1}{10} \times \frac{1}{9}$

8. $\frac{2}{9} \times \frac{3}{10}$

MULITIPLICATION OF MIXED NUMBERS

Note:

-Change mixed number to improve fractions then multiple numerator by numerator and denominator by denominator.

Example 1

Simplify: $1\frac{1}{2}$ x $\frac{1}{3}$

Soln

$1\frac{1}{2}$ x $\frac{1}{3}$

$= \frac{3}{2}$ x $\frac{1}{3}$

$= \frac{1}{2}$

Example II

Simplify $5\frac{1}{5}$ x $3\frac{1}{3}$

Soln

$5\frac{1}{5}$ x $3\frac{1}{3}$ $= \frac{26}{5}$ x $\frac{10}{3}$

$= \frac{26}{5}$ x $\frac{10}{3}$

$= 11$ rem 1

52

3

$= 17\frac{1}{3}$

Activity

Simplify the following

1. $1\frac{1}{2}$ x $1\frac{1}{2}$ 2. $1\frac{1}{4}$ x $\frac{1}{5}$ 3. $1\frac{1}{4}$ x $\frac{1}{5}$ 4. $1\frac{1}{4}$ x $1\frac{1}{5}$
5. $1\frac{1}{3}$ x $2\frac{1}{5}$ 6. $2\frac{1}{4}$ x $1\frac{1}{3}$

MULTIPLICATION OF FRACTIONS USING "OF"

NOTE:

The word of can be replaced by X a multiplication symbol (X)

Example I

Simplify: $\frac{1}{2}$ of 16

$\frac{1}{2}$ of 16 $= \frac{1}{2} \times \frac{16}{1}^{8}$

$= 1 \times 8$

$= 8$

Example II

Simplify: $\frac{2}{3}$ of 10

Soln

$\frac{2}{3}$ of 10 $= \frac{2}{3} \times 10$

$= \frac{2 \times 10}{3}$

$= \frac{20}{3}$ 6rem 2

$= 6\frac{2}{3}$

Example III

What is $\frac{2}{3}$ of $\frac{5}{6}$

Soln

$\frac{2}{3}$ of $\frac{5}{6} = \frac{2}{3} \times \frac{5}{6}$

$= \frac{\cancel{2} \times 5}{3 \times \cancel{6}_{3}}$

$= \frac{1 \times 5}{3 \times 3}$

$= \frac{5}{9}$

Example IV

Simplify: $4\frac{5}{6}$ of 30

Soln

$4\frac{5}{6}$ of 30 = $\frac{29}{6}$ x 30

= $\frac{29}{\cancel{6}}$ x $\frac{\cancel{30}5}{1}$

= 145

Activity

Simplify the following

(1) $\frac{1}{2}$ of 30 (2) $\frac{2}{3}$ of 12 (3) $2\frac{1}{4}$ of 24

(4) $\frac{2}{3}$ of $\frac{3}{8}$ (5) $\frac{1}{4}$ of $\frac{1}{3}$ (6) $\frac{2}{3}$ of $\frac{3}{10}$

(7) $1\frac{5}{7}$ of $3\frac{1}{2}$ (8) $2\frac{2}{5}$ of $3\frac{3}{4}$ (9) $1\frac{5}{7}$ of $3\frac{1}{2}$

APPLICATION OF MULTIPLICATION OF FRACTIONS

Example 1

What is $\frac{1}{4}$ of an hour

Soln

NB: Change 1 hr to minutes

1 hour = 60min

= $\frac{1}{4}$ of 60 minutes

$\frac{1}{4}$ of 1 hour = $\frac{1}{4}$ x 60 minutes

= $\frac{1}{\cancel{4}1}$ x $\frac{\cancel{60}\ 15}{1}$

= 15 minutes

Example II

What is $\frac{2}{3}$ of 18

Soln

$\frac{2}{3}$ of 18 $= \frac{2}{3} \times \frac{18^{6}}{3_{1}} = 2 \times 6 = 12$

Example III

A man spent $\frac{3}{4}$ of his salary on food, if his salary was sh.120,000. How much did he spend on food?

Food $= \frac{3}{4} \times$ sh 120,000

$= \frac{3}{4} \times$ sh120,000

$=$ sh 3×30000

$=$ sh 90,000

Activity

1. What is $\frac{1}{6}$ of 24 kg?
2. What is $\frac{3}{4}$ of sh 400?
3. Find $\frac{1}{4}$ of 1 kg.
4. Tom read $\frac{3}{5}$ of his book that contains 500 pages, how many pages did he read?
5. Mandela covered $\frac{3}{7}$ of 280km by a taxi, how many km did he cover by a taxi?
6. In a village of 1800 people $\frac{1}{9}$ of them are children, how many children are on the village?
7. $\frac{2}{5}$ of my salary is spent on rent. If my salary is sh200,000, how much do I spend on rent

FINDING RECIPROCALS / MULTIPLICATIVE INVERSE

A reciprocal is fraction that when multiplied by a given number the result in one (1)

Example 1

Find the reciprocal of $\frac{2}{3}$

Soln

Let the reciprocal be K

$2 \times K = 1$

$\frac{2K}{3} = 1$

$2K \times 3 = 1 \times 3$

3

$2K = 3$

$\frac{2K}{2} = \frac{3}{2}$

$K = \frac{3}{2}$

The reciprocal of $\frac{2}{3}$ is $\frac{3}{2}$

OR

$\text{Reciprocal} = 1 \div \frac{2}{3}$

$= 1 \times \frac{3}{2}$

$= \frac{3}{2}$

Example II

Find the reciprocal of $2\frac{1}{3}$

Soln

Let the reciprocal be K

$2\frac{1}{3} \times K = 1$

$\frac{7}{3} K = 1$

$\frac{7}{3} K = 1 \times 3$

$7K = 3$

$\frac{7K}{7} = \frac{3}{7}$

$K = \frac{3}{7}$

Example III

What is the reciprocal of 5?

Let the reciprocal be P

$5 \times P = 1$

$5P = 1$

$\frac{5P}{5} = \frac{1}{5}$

$P = \frac{1}{5}$

Example IV

Find the reciprocal of 0.6

Let the reciprocal be y

$0.6 \times y = 1$

$\frac{6}{10} \times y = 1$

$\frac{6y}{10} = 1$

$\frac{6y}{10} \times 10 = 1 \times 10$

$6y = 10$

$\frac{6y}{6} = \frac{10}{6}$

$1 \quad 6$

$Y = \frac{10}{6}$

Activity

Find the reciprocal of the following.

(1) $\frac{3}{4}$ (4) $2\frac{1}{2}$ (7) 4 (10) 0.3

(2) $\frac{4}{7}$ (5) $3\frac{1}{4}$ (8) 12 (11) 1.4

(3) $\frac{4}{9}$ (6) $4\frac{1}{5}$ (9) 7 (12) 0.9

DIVISION OF FRACTIONS BY WHOLES AND WHOLES BY FRACTIONS.

Steps to take:

Maintain the first fraction, change the division sign to multiplication sign then reciprocate the second fraction.

Example I

Work out: $\frac{2}{3} \div 2$

Soln

$= \frac{2}{3} \div \frac{2}{1}$

$= \frac{2}{3} \times \frac{1}{2}$

$= \frac{1}{3} \times \frac{1}{1}$

$= \frac{1}{3}$

Example II

Simplify: $3\frac{1}{2} \div 35$

$3\frac{1}{2} \div 35 = \frac{7}{2} \div \frac{35}{1}$

$= \frac{7}{2} \times \frac{1}{35}$

$= \frac{1 \times 1}{2 \times 5}$

$= \frac{1}{10}$

Example III

Simplify: $6 \div \frac{1}{4}$

Soln

$6 \div 1 = 6 \div \frac{1}{4}$

$= \frac{6}{1} \times \frac{4}{1}$

$= 24$

Example IV

Divide: $8 \div 2\frac{1}{2}$

Soln

$8 \div 2\frac{1}{2}$

$= 8 \div \frac{5}{2}$

$= \frac{8}{1} \times \frac{2}{5}$

$= \frac{16}{5}$ 3 rem 1

$= 3\frac{1}{5}$

Activity

Work out the following

a) $\frac{2}{5} \div 2$

b) $4 \div \frac{2}{3}$

c) $\frac{1}{3} \div 3$

d) $16 \div \frac{4}{3}$

e) $\frac{4}{5} \div 3$

f) $9 \div \frac{4}{5}$

DIVISION OF WHOLES AND BY FRACTIONS USING REPEATED ADDITION.

Example

Divide $2 \div \frac{1}{4}$ using repeated addition.

Step1

Find the equivalent fraction of 2 whose denominator is 4.

$$\frac{2 \times 4}{1 \times 4} = \frac{8}{4}$$

Step2

Divide $\frac{8}{4}$ using repeated subtraction

$$2 \div \frac{1}{4} \longrightarrow \frac{8}{4} \div \frac{1}{4}$$

$\frac{8}{4} - \frac{1}{4} = \frac{7}{4}$ 1st

$\frac{7}{4} - \frac{1}{4} = \frac{6}{4}$ 2nd

$\frac{6}{4} - \frac{1}{4} = \frac{5}{4}$ 3rd

$\frac{5}{4} - \frac{1}{4} = \frac{4}{4}$ 4th

$\frac{4}{4} - \frac{1}{4} = \frac{3}{4}$ 5th

$\frac{3}{4} - \frac{1}{4} = \frac{2}{4}$ 6th

$\frac{2}{4} - \frac{1}{4} = \frac{1}{4}$ 7th

$\frac{1}{4} - \frac{1}{4} = 0$ 8th

$2 \div \frac{1}{4} = 8$

Activity.

Work out the following using repeated subtraction

(i) $1 \div \frac{1}{4}$

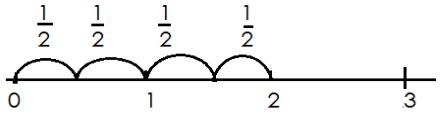
(ii) $5 \div \frac{1}{2}$

(iii) $2 \div \frac{2}{3}$

DIVISION OF WHOLES AND BY FRACTIONS USING A NUMBERLINE.

Example.

Divide $2 \div \frac{1}{2}$ using a numberline.



$2 \div \frac{1}{2} = 4$

Activity.

(c) Work out the following using a number line

(a) $2 \div \frac{1}{3}$ (b) $8 \div \frac{3}{4}$ (c) $2 \div \frac{1}{4}$

Division of fraction by fractions

Example

Divide: $\frac{1}{3} \div \frac{1}{4}$

Soln

$\frac{1}{3} \div \frac{1}{4} = \frac{1}{3} \times \frac{4}{1}$

$= \frac{1 \times 4}{3 \times 1}$

$= \frac{4}{3}$ 1rem 1

$= 1\frac{1}{3}$

Example II

Simplify: $\frac{4}{5} \div \frac{2}{15}$

$= \frac{4}{5} \div \frac{2}{15}$

$= \frac{4}{5} \times \frac{15}{2}$

$= 6$

Example II

Simplify: $\frac{1}{2} \div 2\frac{1}{2}$

Soln.

$= \frac{1}{2} \div 2\frac{1}{2}$

$= \frac{1}{2} \times \frac{2}{5}$

$= \frac{1}{5}$

Example IV

Simplify: $2\frac{1}{3} \div 3\frac{1}{2}$

Soln

$= \frac{7}{3} + \frac{7}{2}$

$= \frac{7}{3} + \frac{2}{7}$

$= \frac{1}{3} \times \frac{2}{1}$

$= \frac{2}{3}$

Activity.

Work out the following

(1) $\frac{1}{2} \div \frac{1}{8}$ **(4)** $\frac{1}{5} \div \frac{1}{4}$ **(7)** $\frac{1}{4} \div 4\frac{1}{4}$

(2) $\frac{3}{4} \div \frac{1}{2}$ **(5)** $\frac{1}{2} \div 2\frac{1}{3}$ **(8)** $2\frac{1}{4} \div 1\frac{1}{4}$

(3) $\frac{1}{3} \div \frac{1}{6}$ **(6)** $\frac{1}{3} \div 2\frac{1}{3}$ **(9)** $2\frac{1}{4} \div 1\frac{1}{4}$

(10) $2\frac{1}{2} \div \frac{1}{2}$

APPLICATION OF FRACTIONS INVOLVING DIVISION

Example I

How many $\frac{1}{3}$ Chapatis can be got from 2 Chapati?

$= 2 \div \frac{1}{3}$

$= \frac{2}{1} \div \frac{1}{3}$

$= \frac{2}{1} \div \frac{1}{3}$

$= \frac{2}{1} \times \frac{3}{1}$

$= 6$

Example II

How many $\frac{1}{4}$ litre bottles of milk can be get from 1 $\frac{1}{2}$ litres

No. of bottle $= 4\frac{1}{2} \div \frac{1}{4}$

$= \frac{9}{2} \div \frac{1}{4}$

$= \frac{9}{2} \times \frac{4}{1}$

$= \frac{9}{2} \times \frac{4^{2}}{1}$

$= 18$ bottles

Example III

How many small packets of $1\frac{1}{2}$ kg can be got from a sack of $9\frac{2}{3}$ kg?

No. of packets $= 9\frac{2}{3} \div 1\frac{1}{3}$

$= \frac{29}{3} \div \frac{4}{3}$

$= \frac{29}{3} \times \frac{3}{4}$

$= \frac{29}{4}$ 7rem 1

$= 7\frac{1}{4}$ packets

Activity

1) How many $\frac{1}{4}$ cakes be got from 3 cakes?

2) If we use $\frac{1}{5}$ kg of sugar per day, how many days shall we use $12\frac{3}{5}$ kg.

3) How many $\frac{1}{2}$ litre bottle can get got forty litre jerry can?

4) A farmer has $7\frac{1}{2}$ hectares of land, to be divided into plots of $1\frac{1}{4}$ hectares, how many plots will he get?

5) A bag contains 5 $\frac{1}{2}$kg of maize. Find to be number of $\frac{1}{2}$kg packets that will be obtained.

6) A woman has 13 $\frac{1}{3}$kg of sugar. How many packets each $\frac{2}{3}$ kg does she have?

7) 12 litres of milk were given to child if each get $\frac{1}{4}$ litre of milk, how many children were server altogether?

END OF TERM 1 WORK